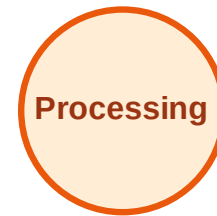


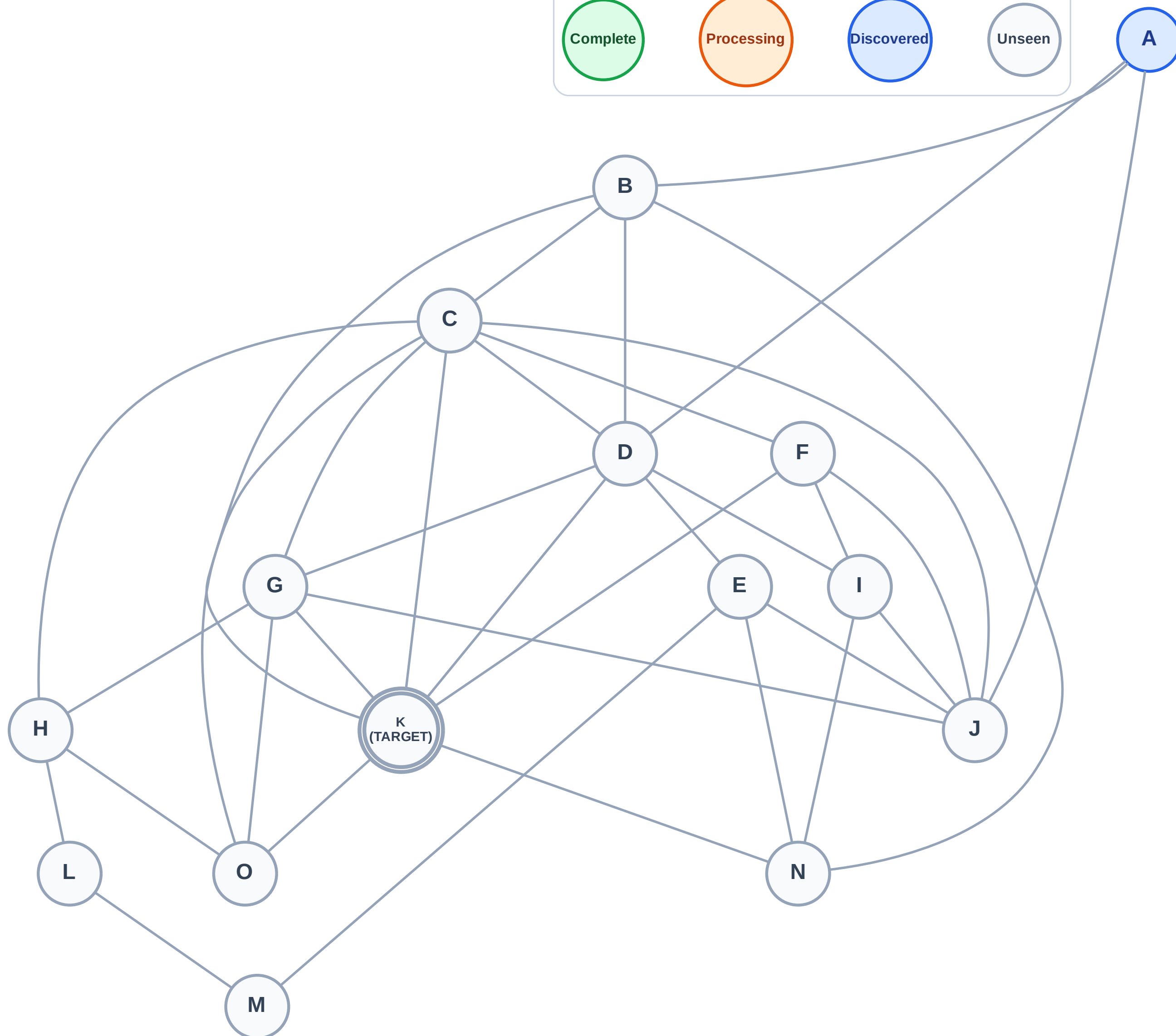
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



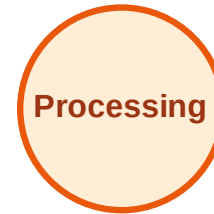
Queue: A
Visited: (none)



BFS Traversal

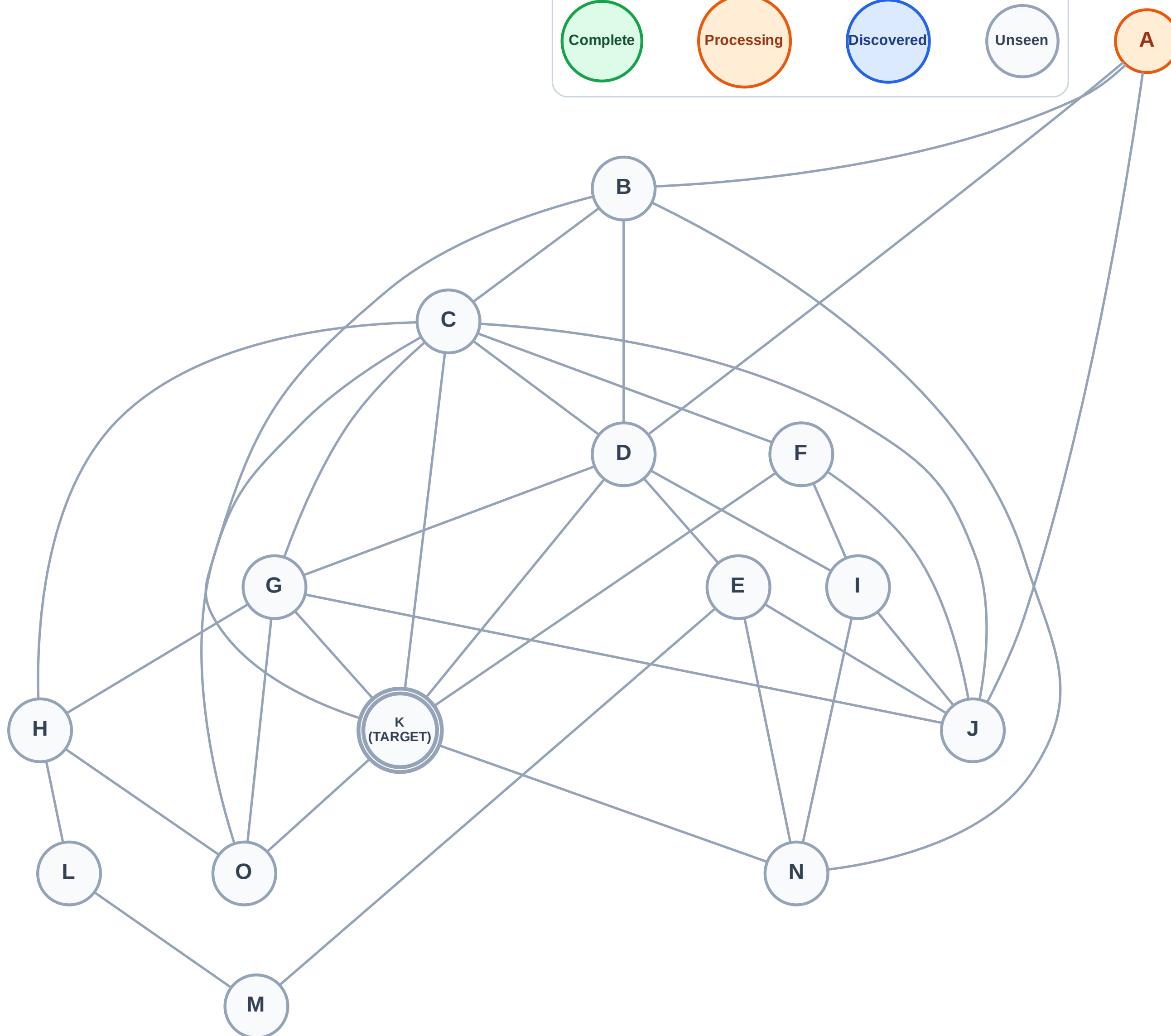
Step 2: Process node A

Legend



Queue: (empty)

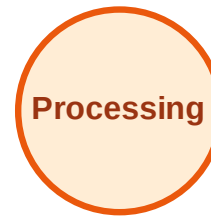
Visited: (none)



BFS Traversal

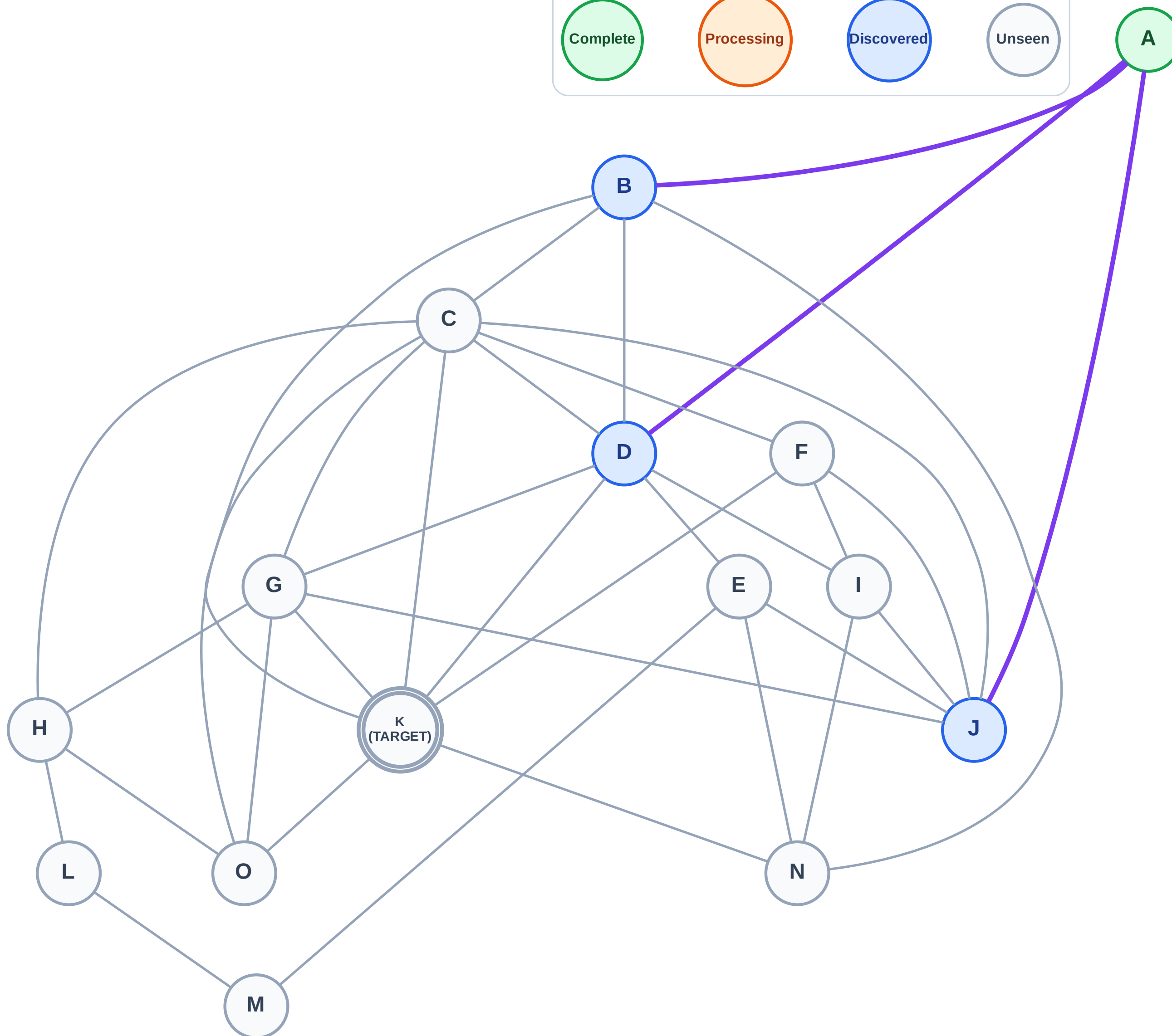
Step 3: Discover B, D, J from A

Legend



Queue: B → D → J

Visited: A



BFS Traversal

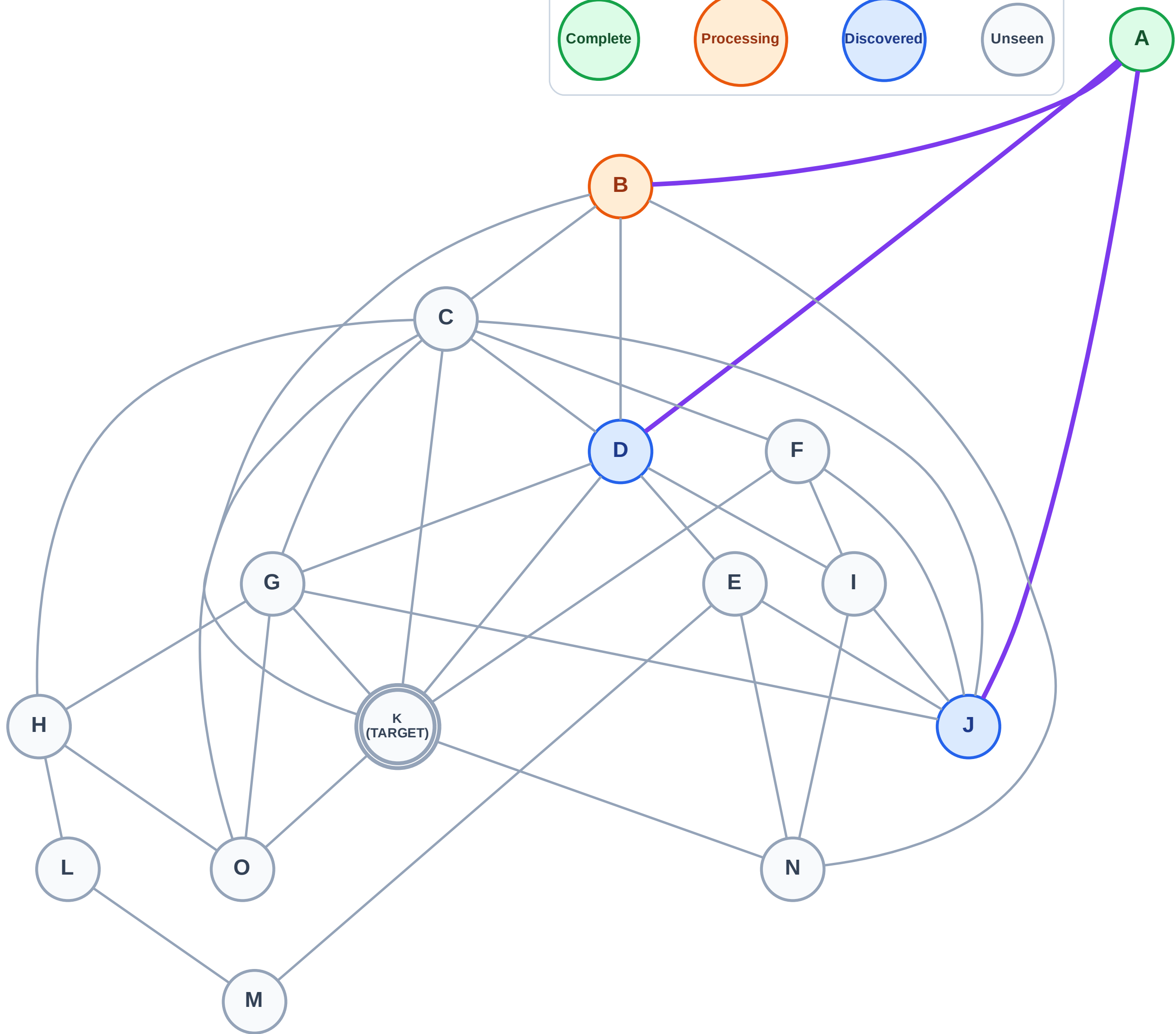
Step 4: Process node B

Legend



Queue: D → J

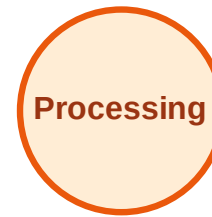
Visited: A



BFS Traversal

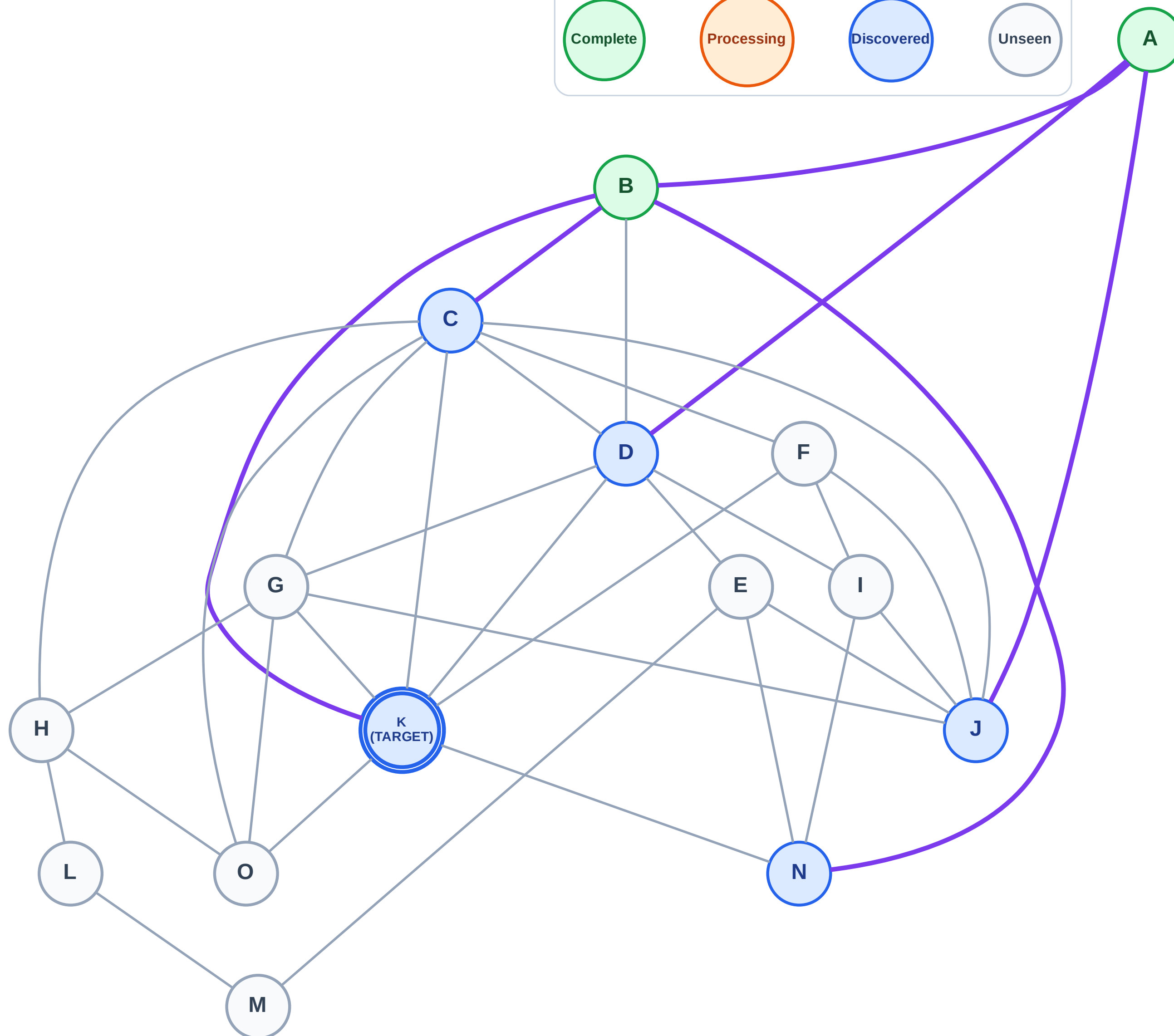
Step 5: Discover C, K, N from B

Legend



Queue: D → J → C → K → N

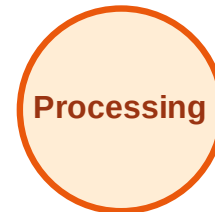
Visited: A, B



BFS Traversal

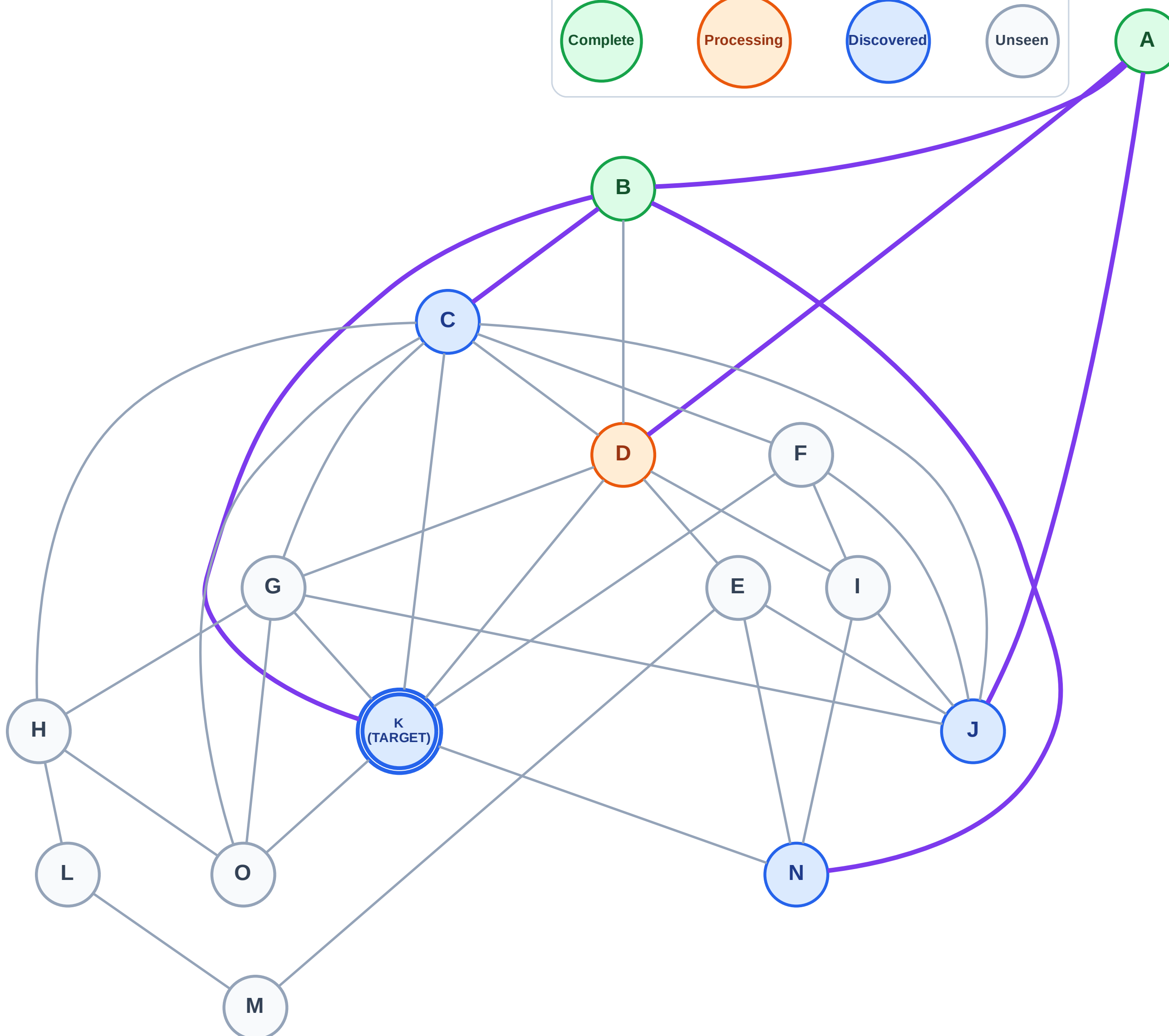
Step 6: Process node D

Legend



Queue: J → C → K → N

Visited: A, B



BFS Traversal

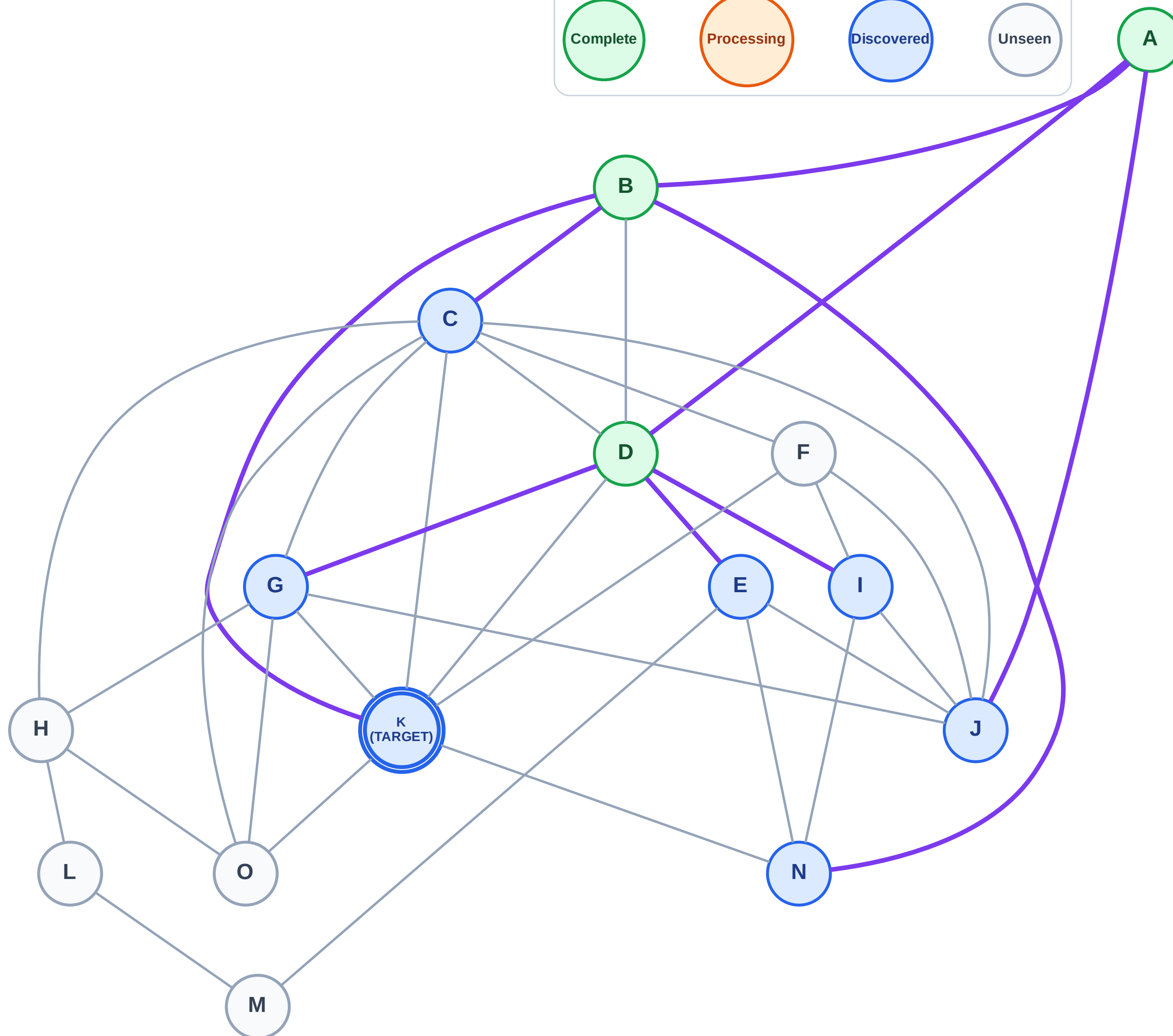
Step 7: Discover E, G, I from D

Legend



Queue: J → C → K → N → E → G → I

Visited: A, B, D



BFS Traversal

Step 8: Process node J

Legend

Complete

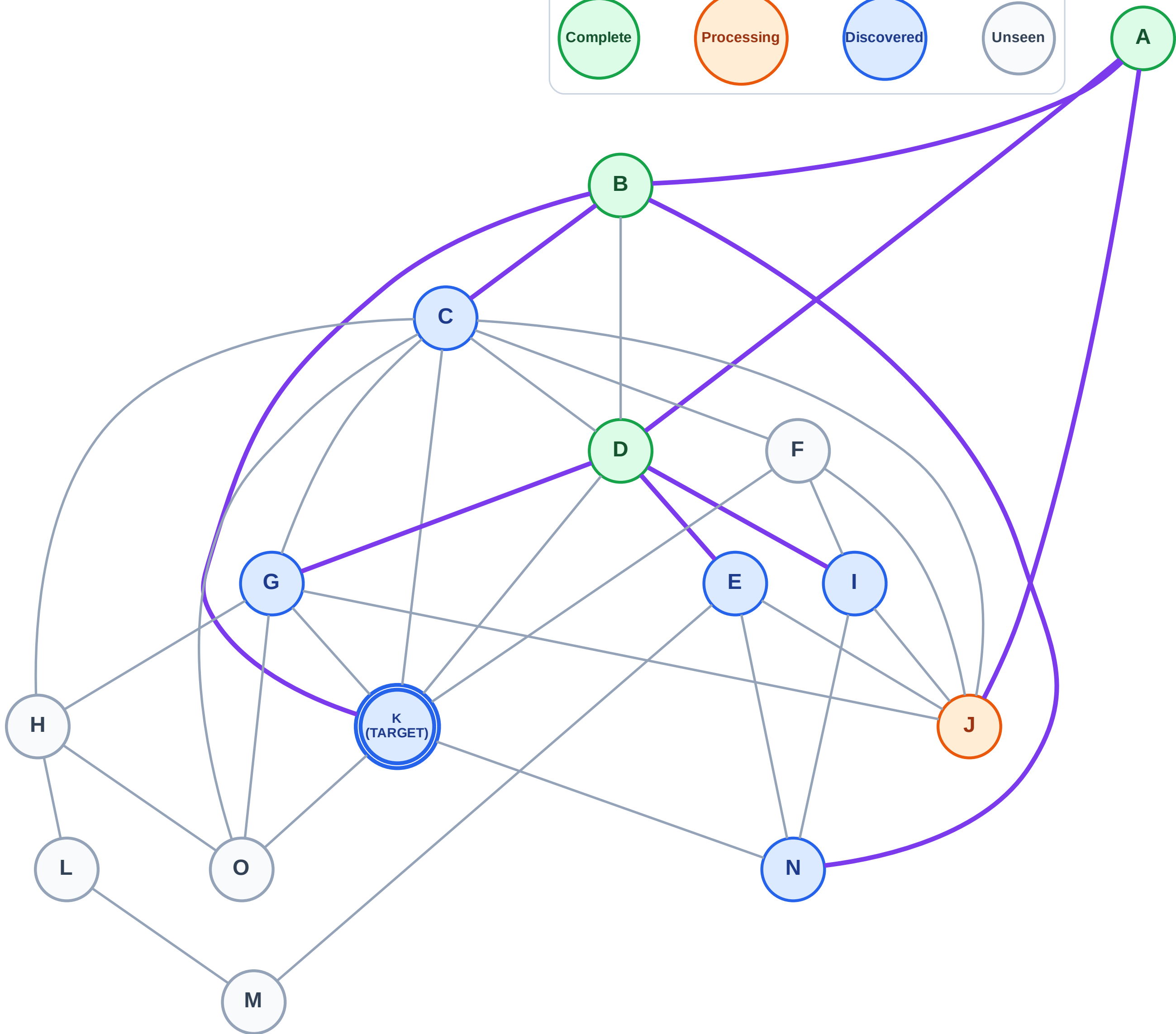
Processing

Discovered

Unseen

Queue: C → K → N → E → G → I

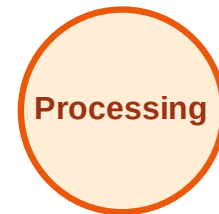
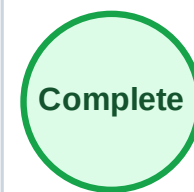
Visited: A, B, D



BFS Traversal

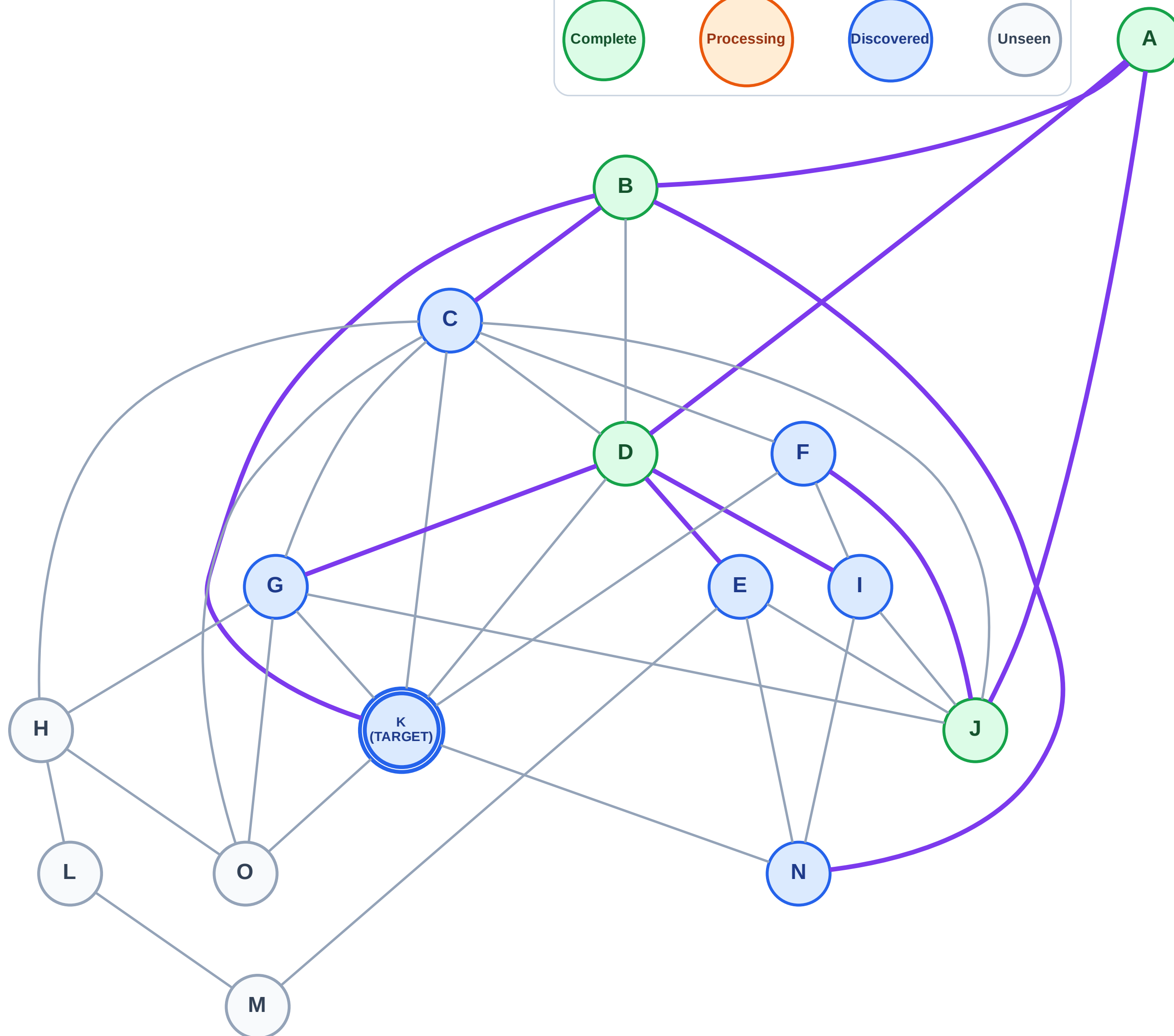
Step 9: Discover F from J

Legend



Queue: C → K → N → E → G → I → F

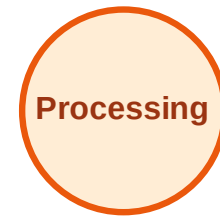
Visited: A, B, D, J



BFS Traversal

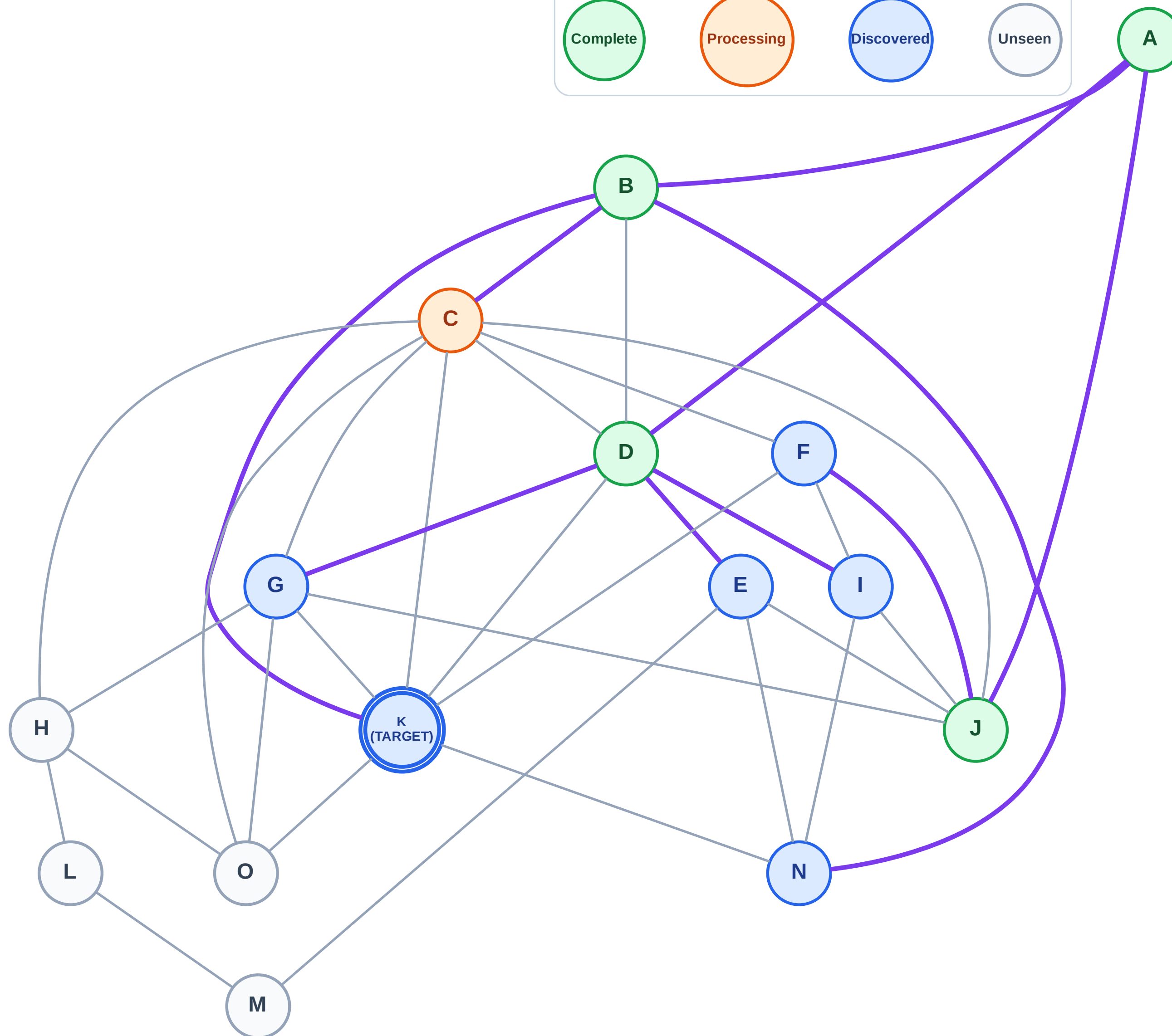
Step 10: Process node C

Legend



Queue: K → N → E → G → I → F

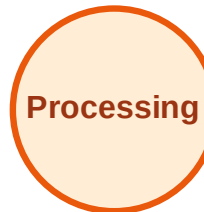
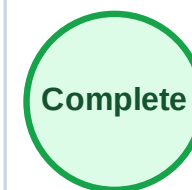
Visited: A, B, D, J



BFS Traversal

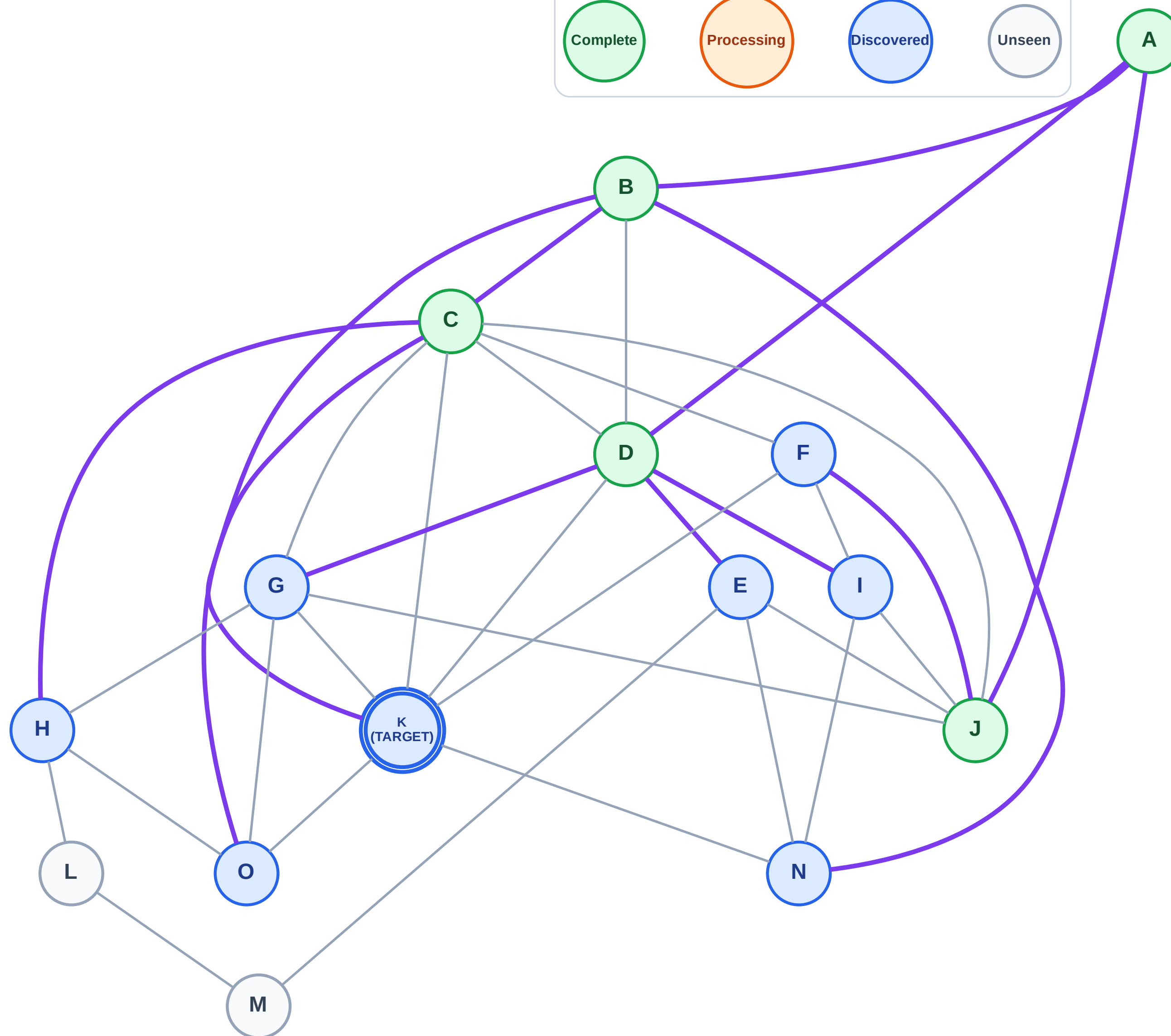
Step 11: Discover H, O from C

Legend



Queue: K → N → E → G → I → F → H → O

Visited: A, B, C, D, J



BFS Traversal

Step 12: Process node K

Legend

Complete

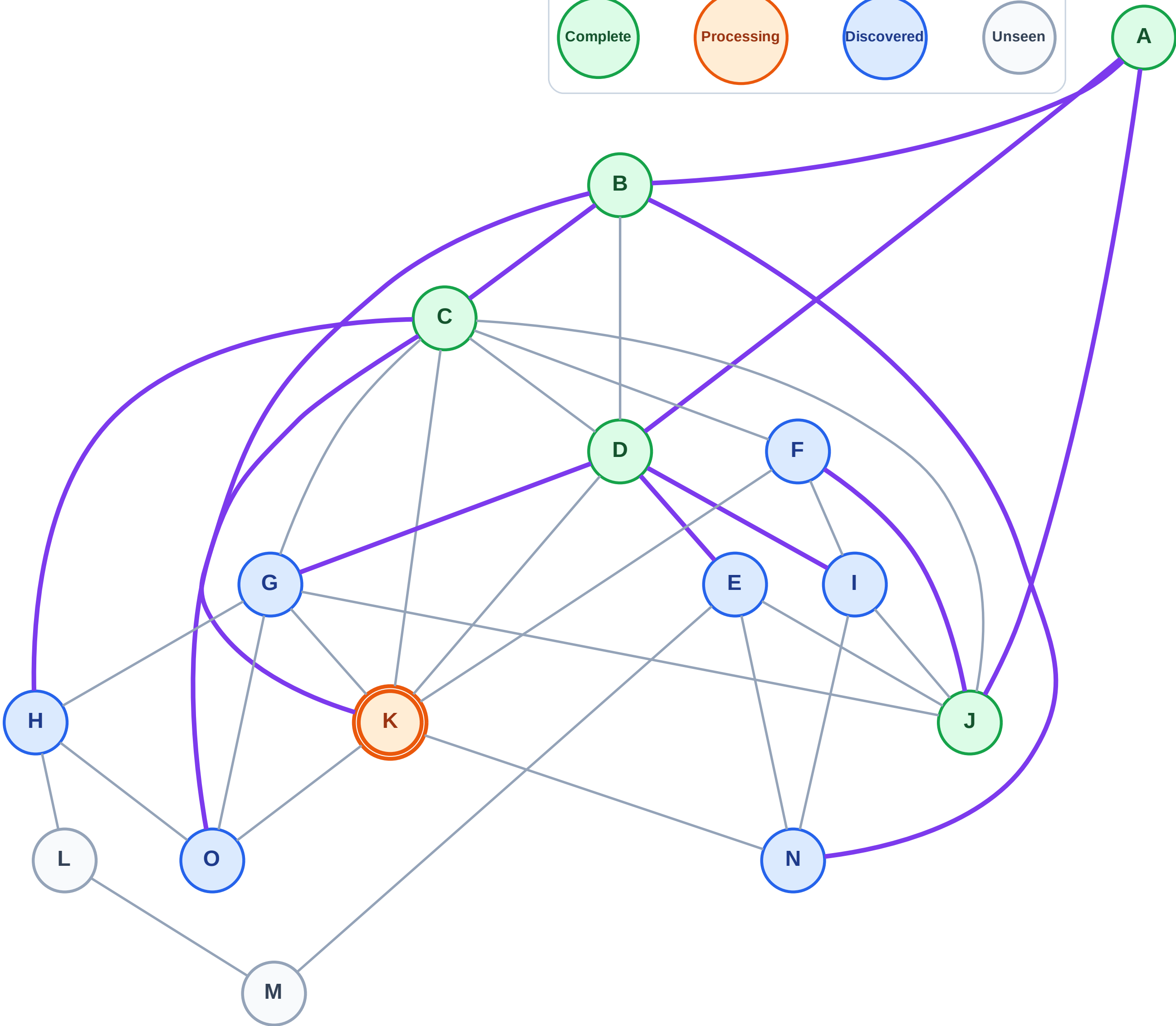
Processing

Discovered

Unseen

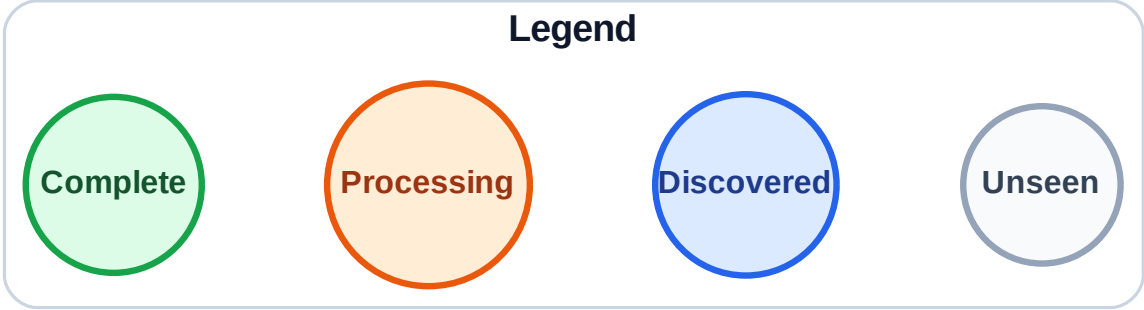
Queue: N → E → G → I → F → H → O

Visited: A, B, C, D, J



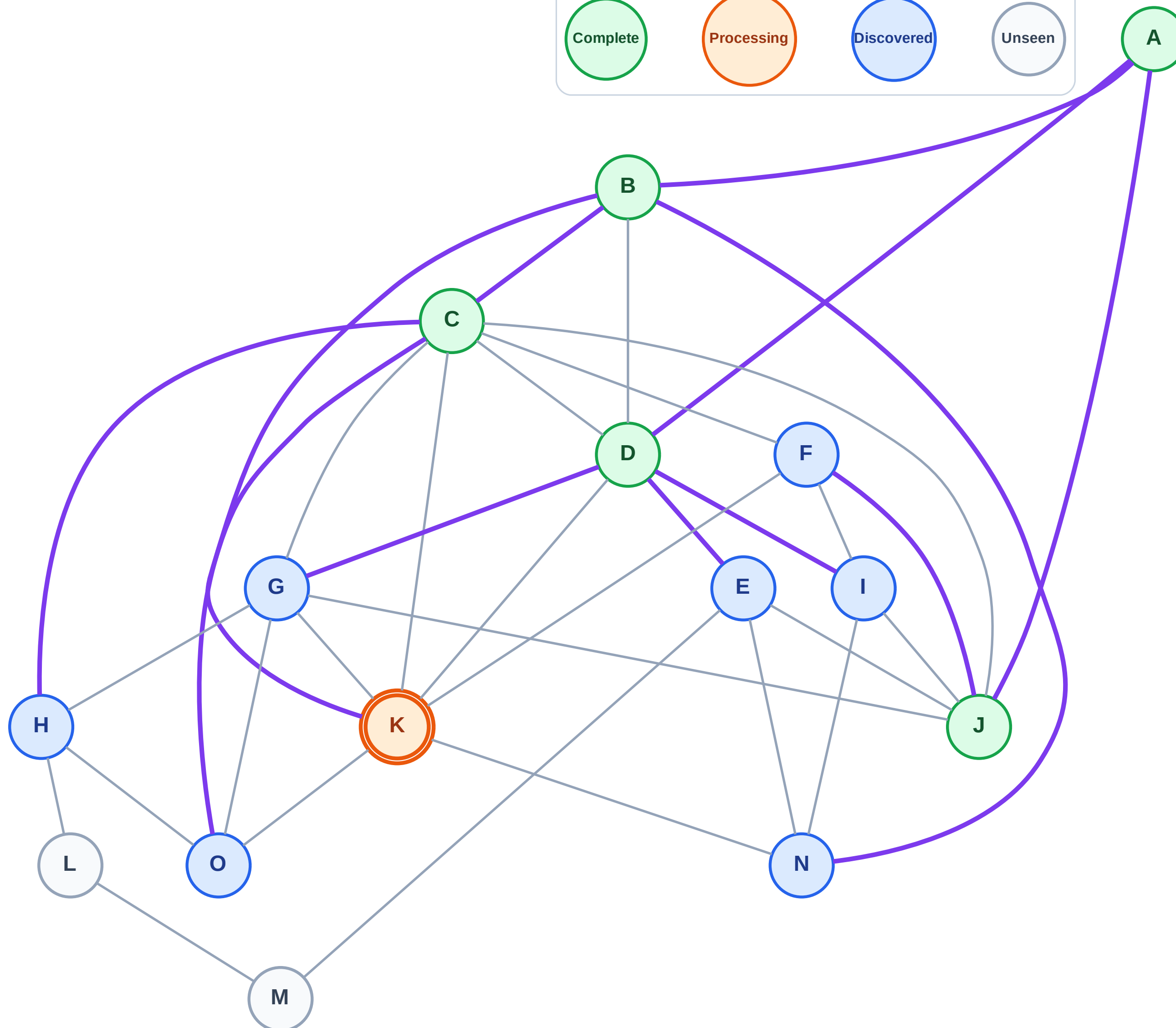
BFS Traversal

Step 13: TARGET NODE K REACHED! Path (2 hops): $A \rightarrow B \rightarrow K$



Queue: N → E → G → I → F → H → O

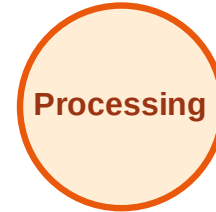
Visited: A, B, C, D, J, K



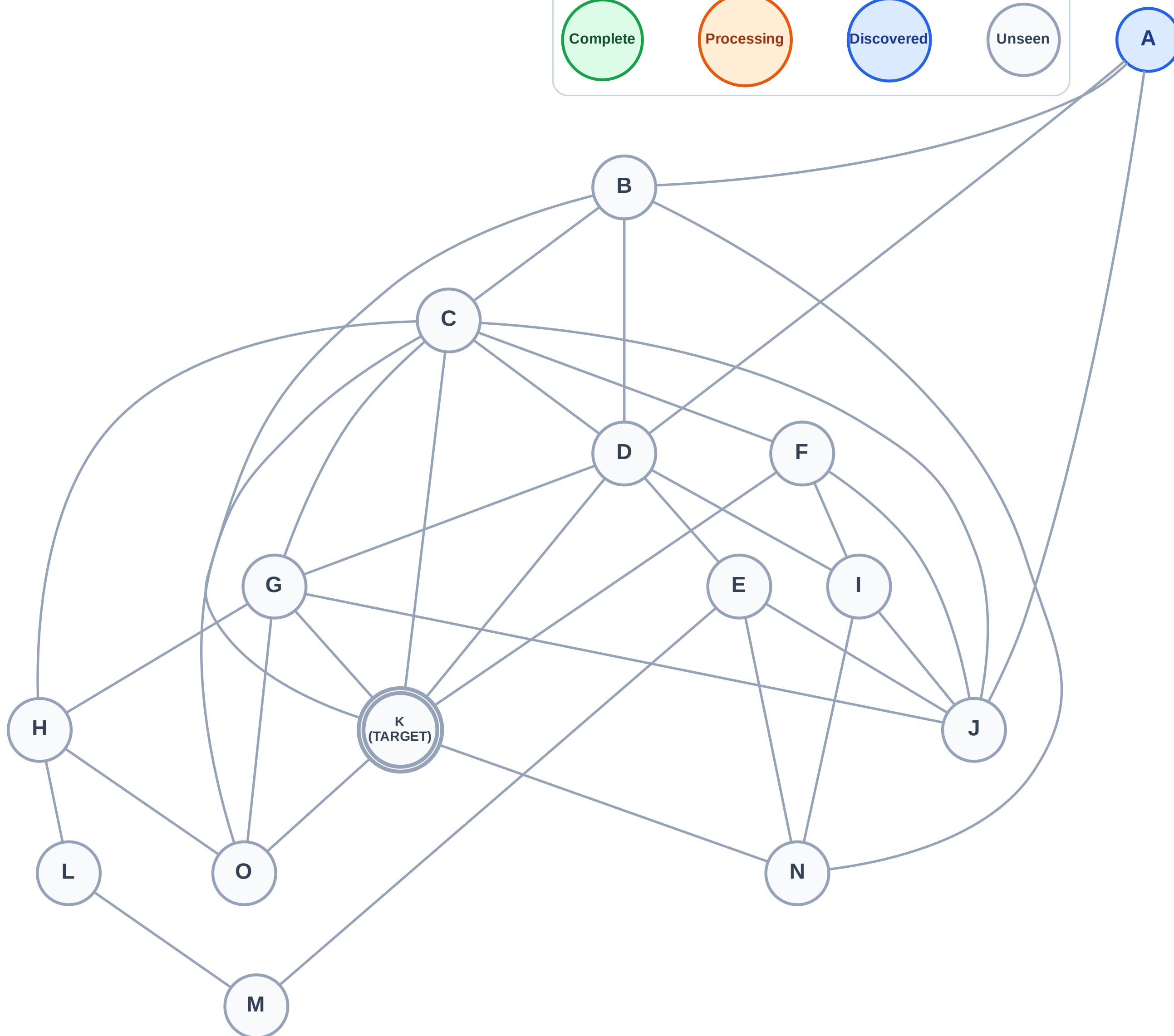
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

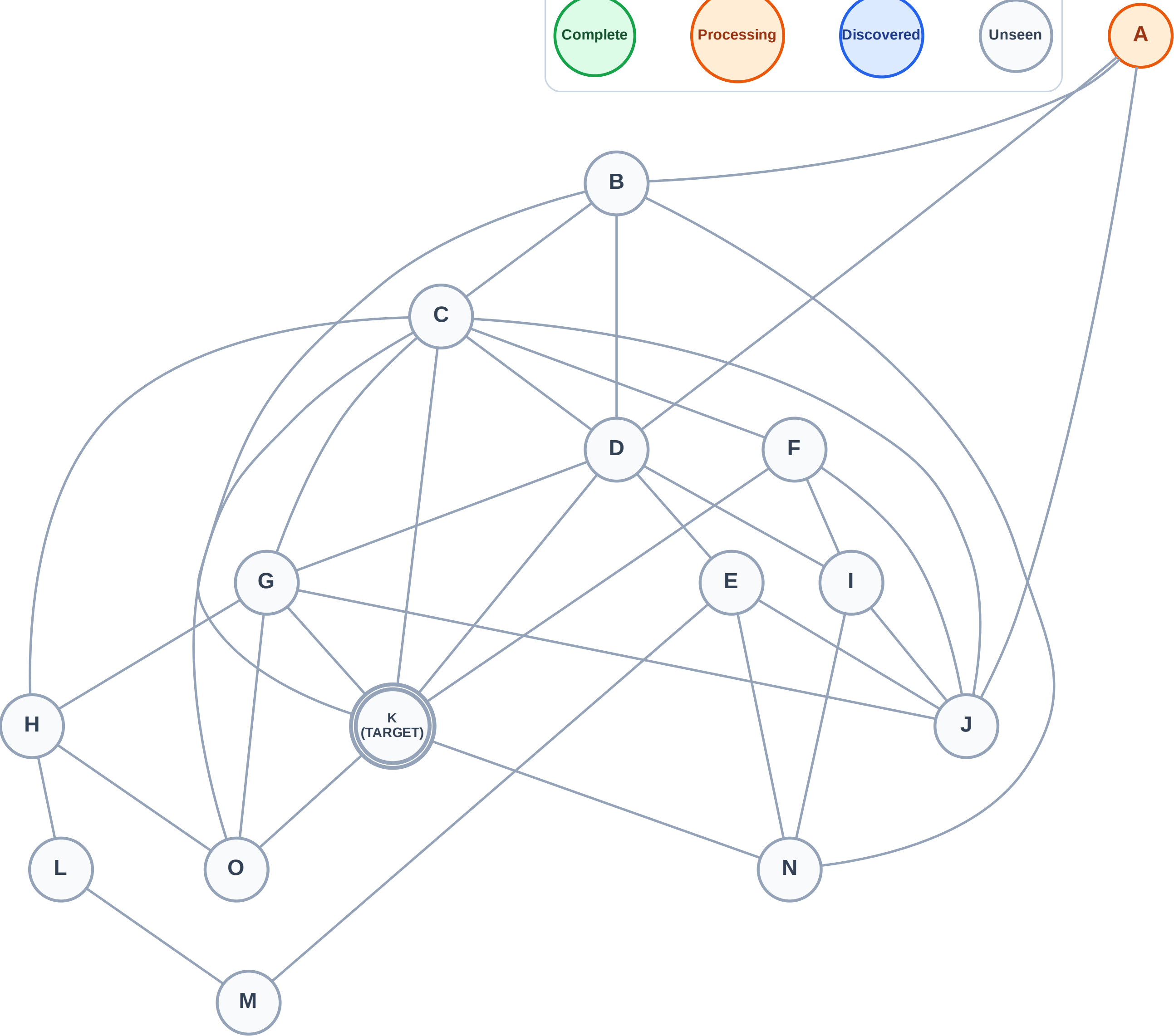
Complete

Processing

Discovered

Unseen

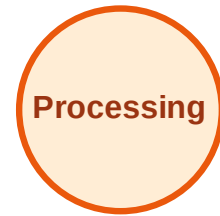
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

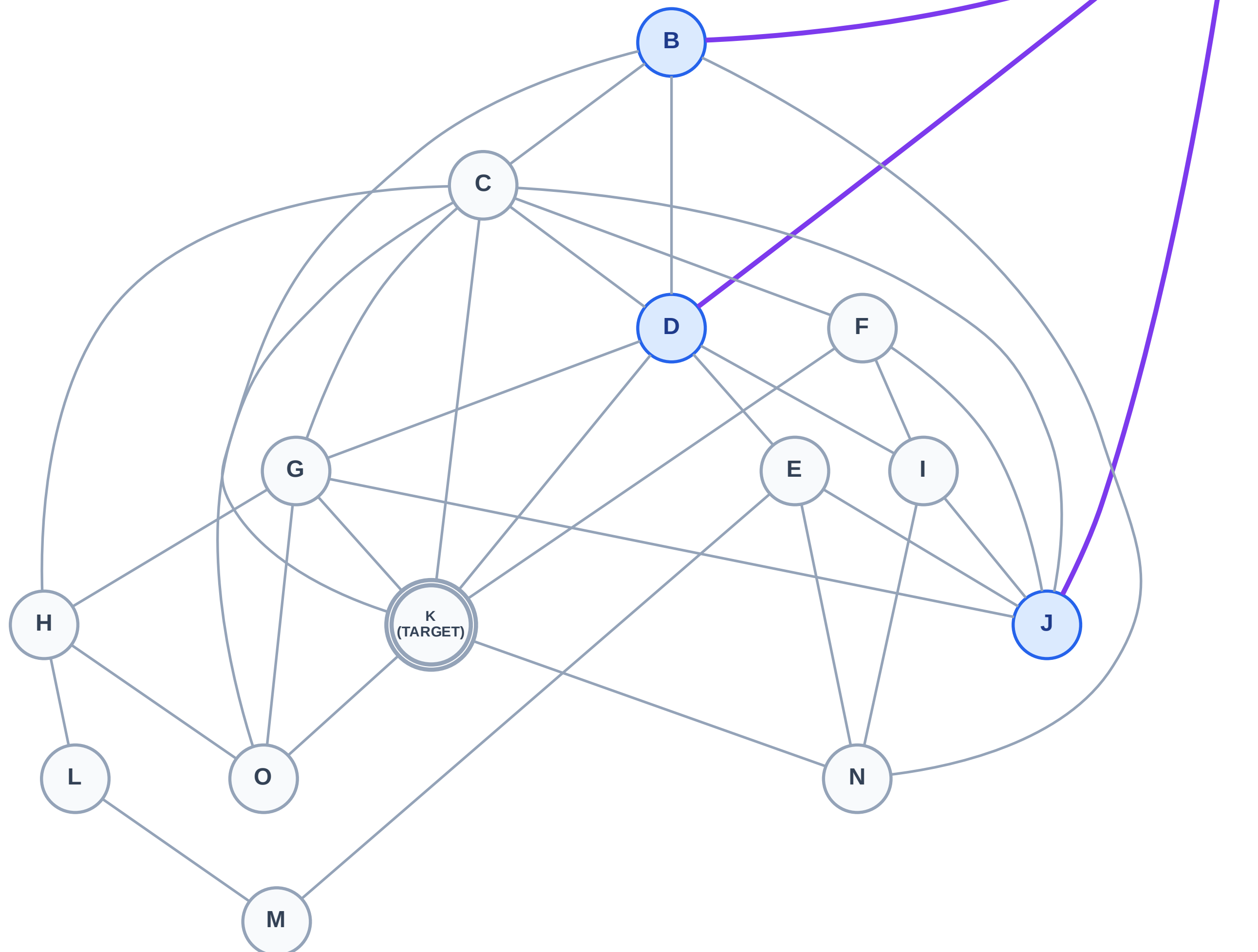
Step 3: Discover J, D, B from A

Legend



Stack (top at right): J | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

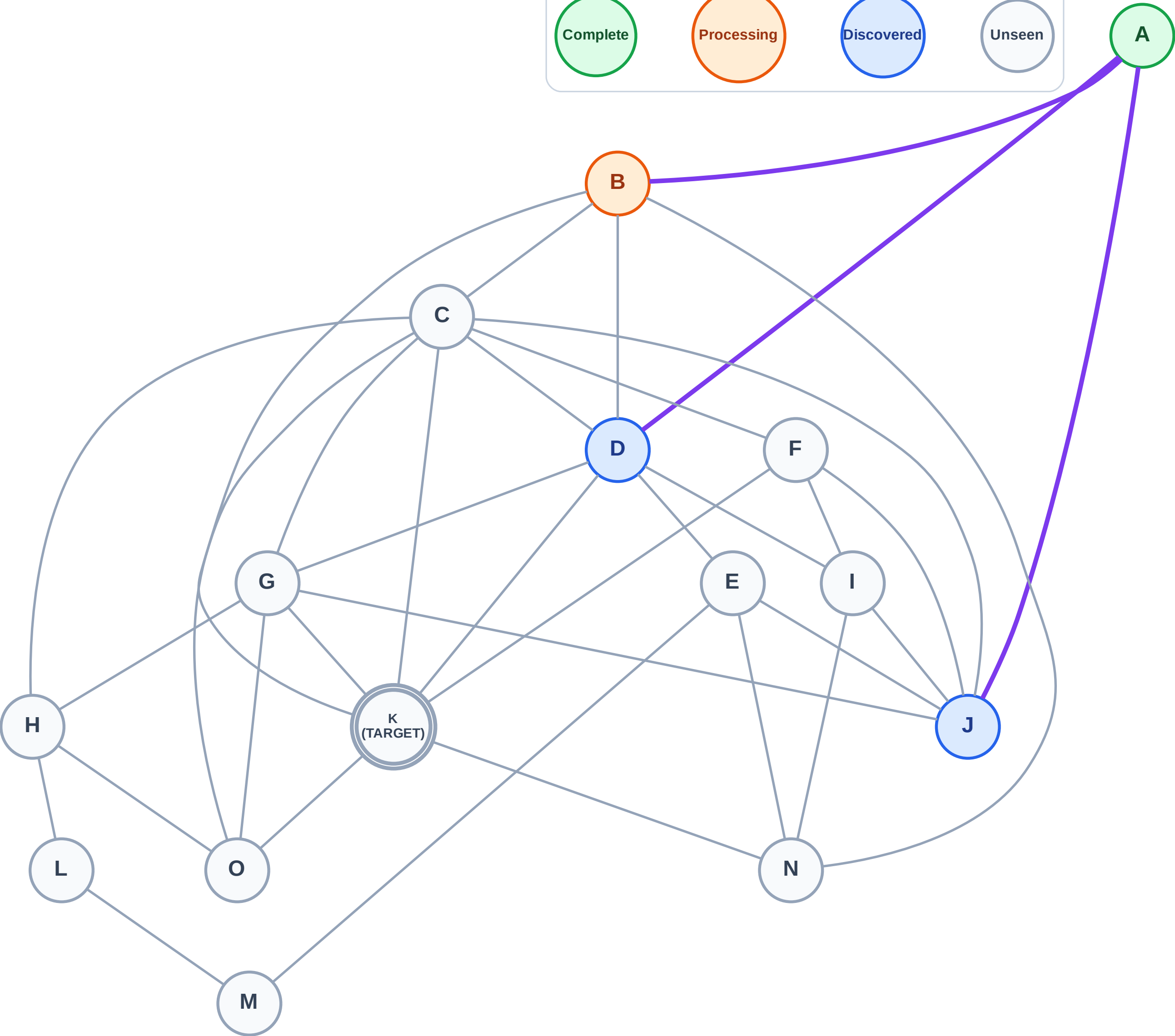
Processing

Discovered

Unseen

Stack (top at right): J | D

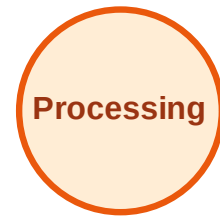
Visited: A



DFS Traversal

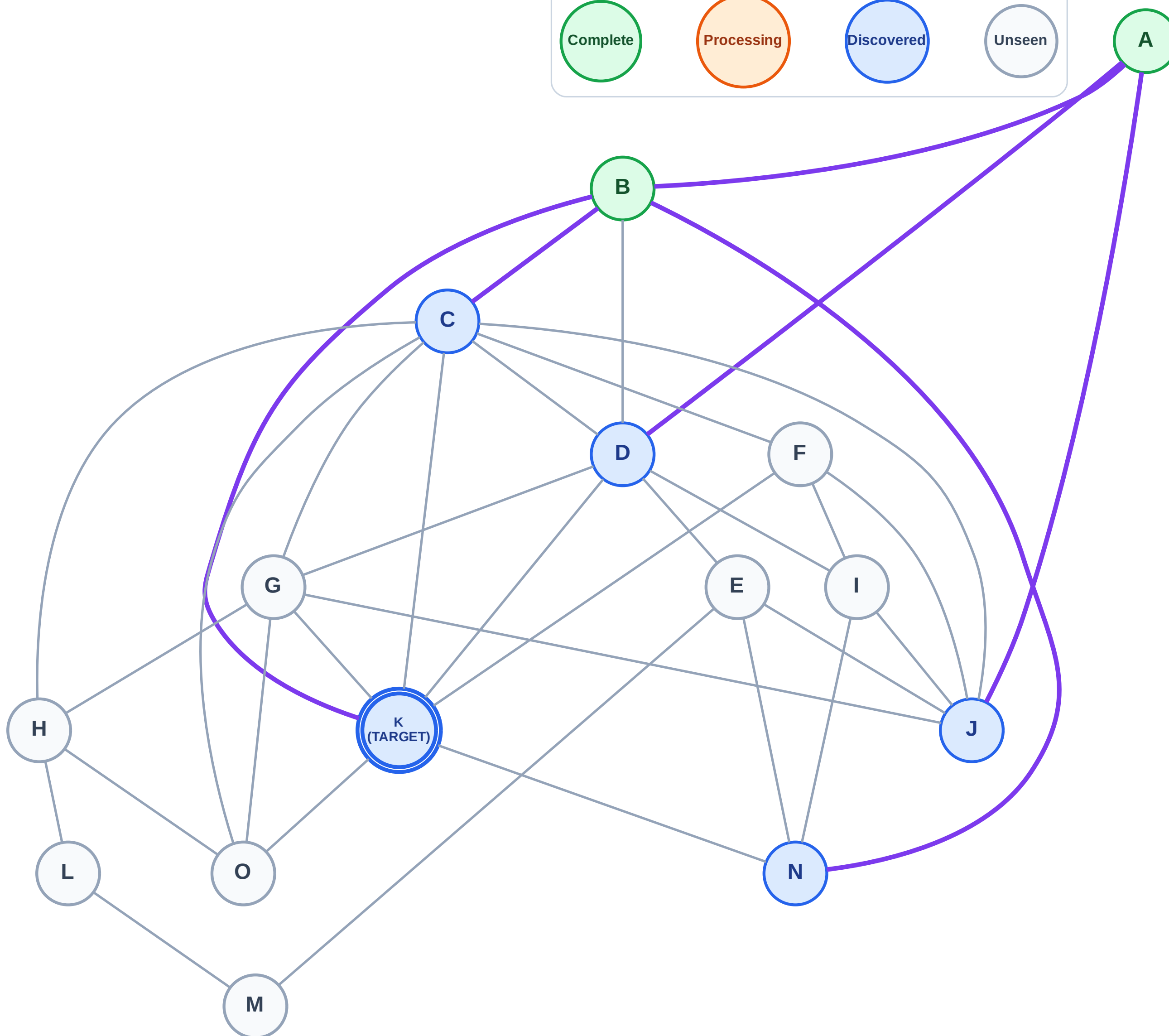
Step 5: Discover N, K, C from B

Legend



Stack (top at right): J | D | N | K | C

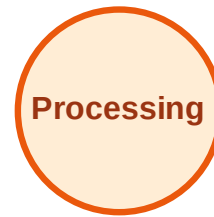
Visited: A, B



DFS Traversal

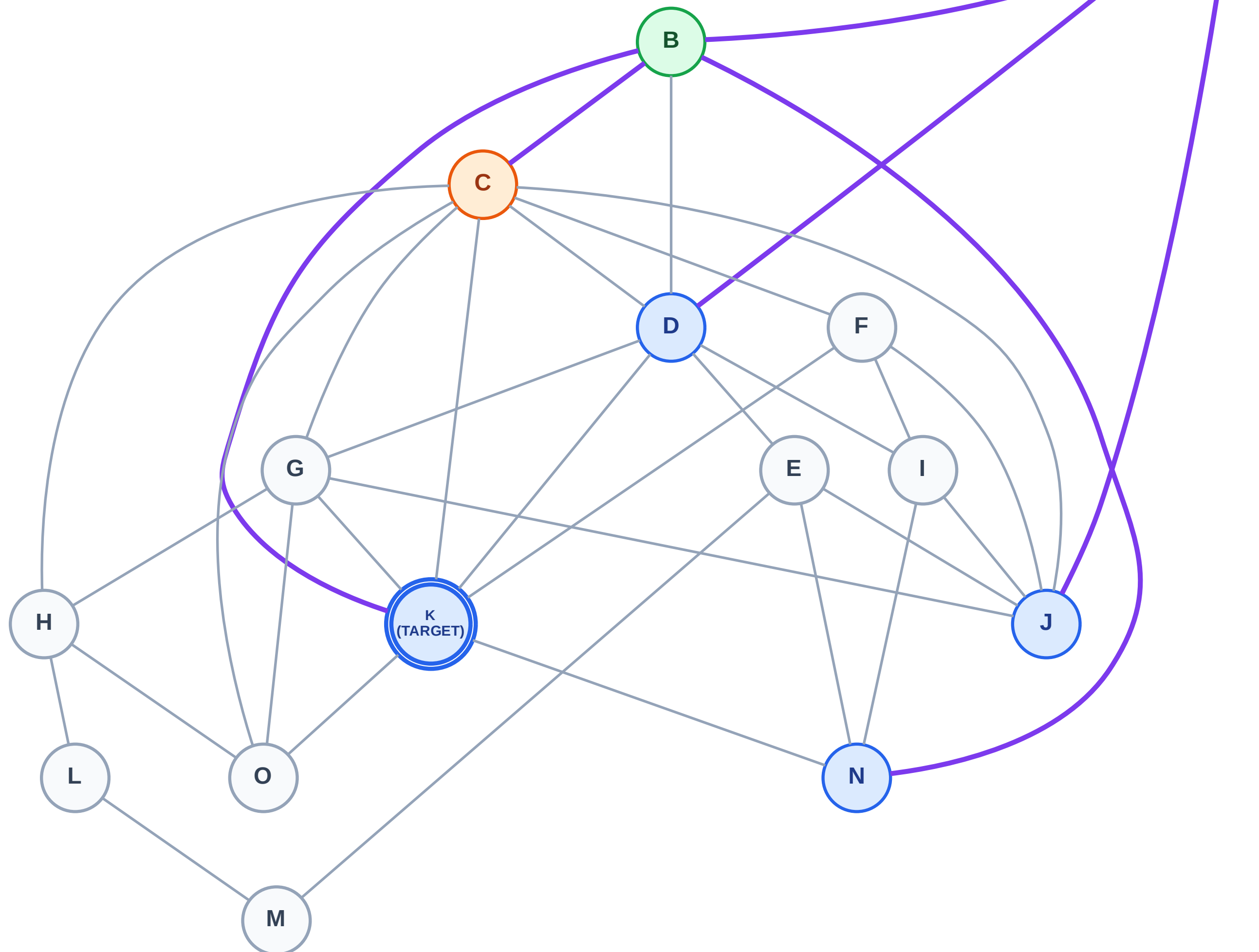
Step 6: Process node C

Legend



Stack (top at right): J | D | N | K

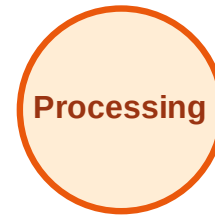
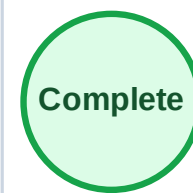
Visited: A, B



DFS Traversal

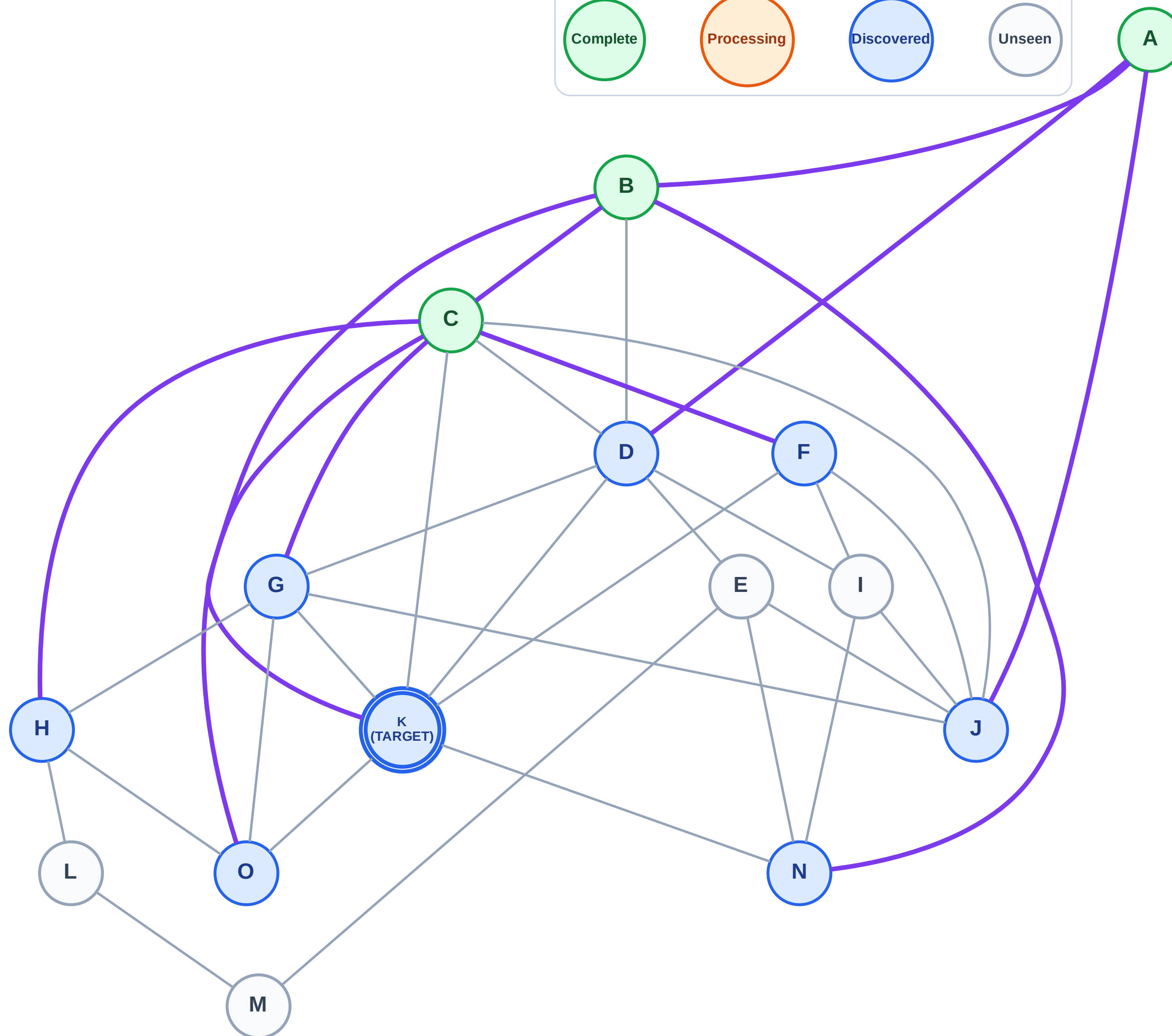
Step 7: Discover O, H, G, F from C

Legend



Stack (top at right): J | D | N | K | O | H | G | F

Visited: A, B, C



Step 8: Process node F

Step 8: Process node F

Legend

Complete

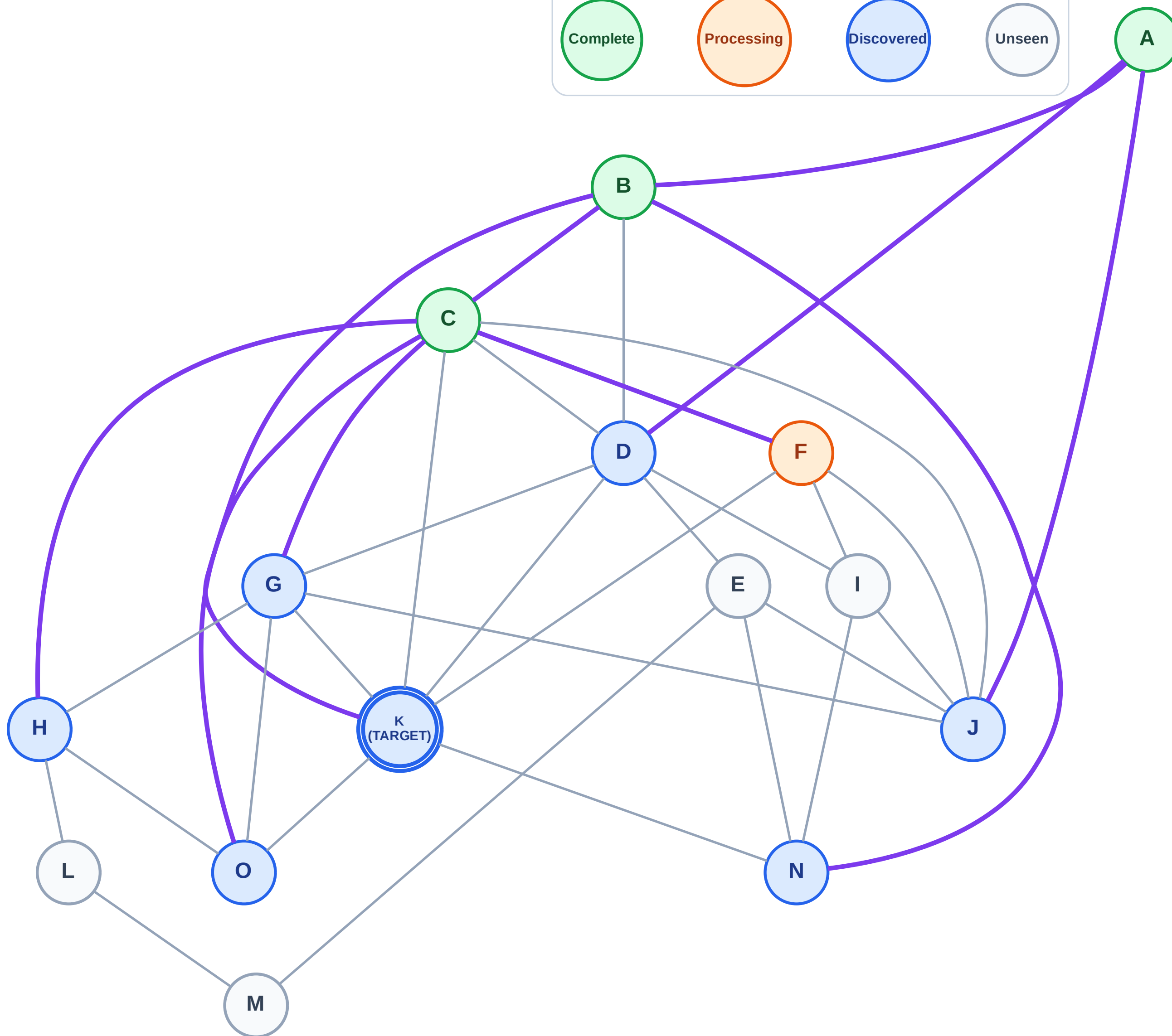
Processing

Discovered

Unseen

Stack (top at right): J | D | N | K | O | H | G

Visited: A, B, C



DFS Traversal

Step 9: Discover I from F

Legend

Complete

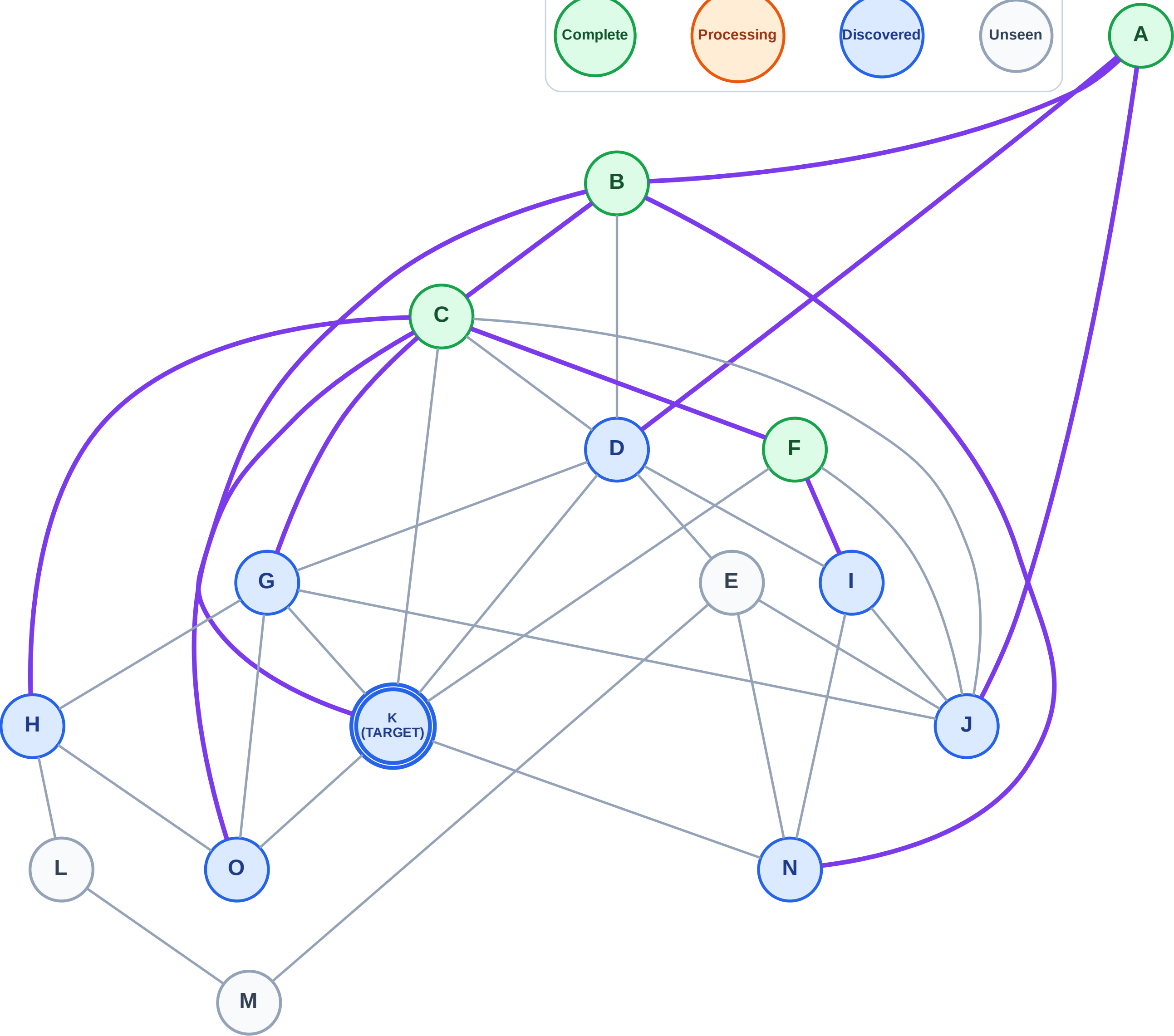
Processing

Discovered

Unseen

Stack (top at right): J | D | N | K | O | H | G | I

Visited: A, B, C, F



DFS Traversal

Step 10: Process node I

Legend

Complete

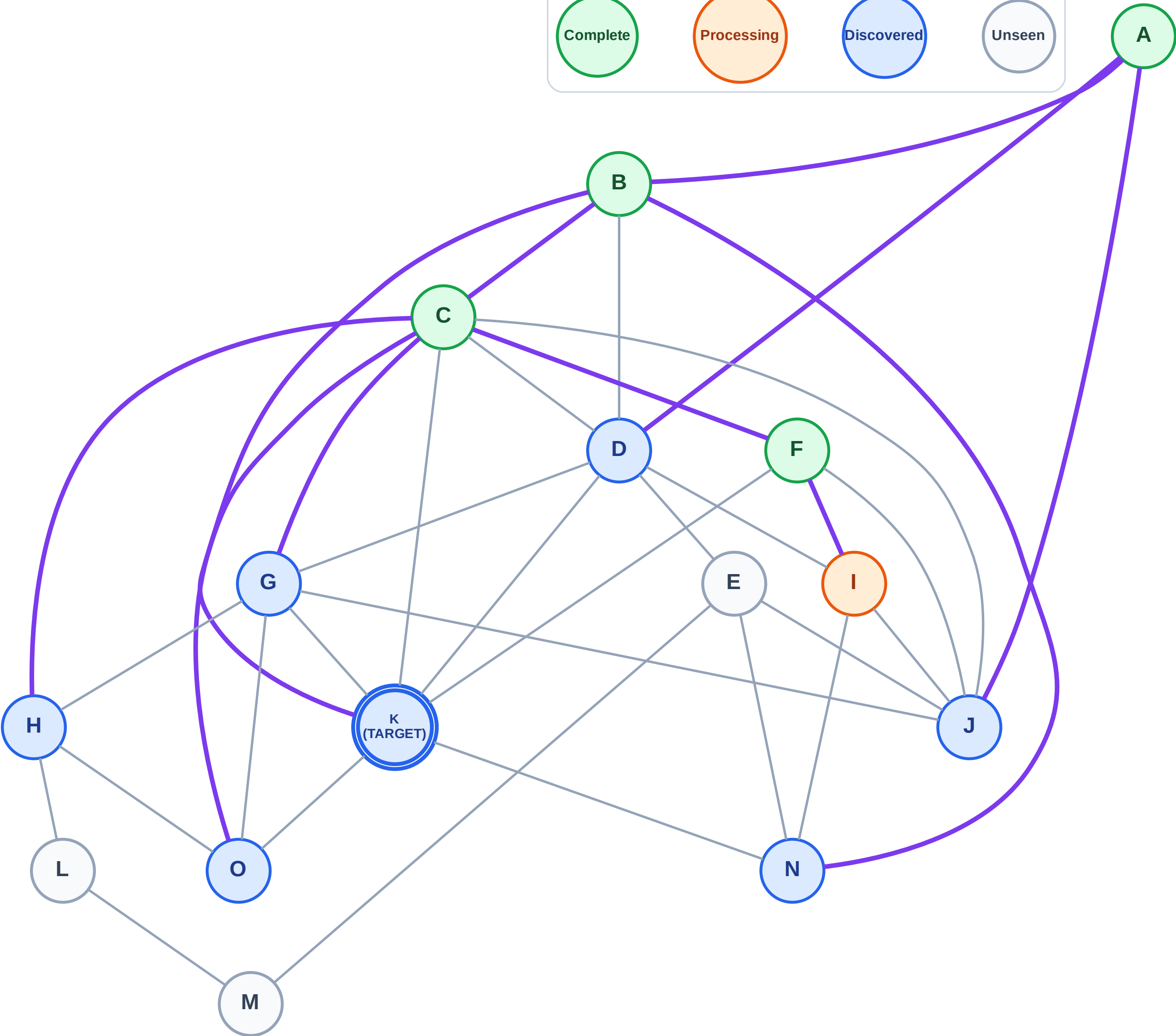
Processing

Discovered

Unseen

Stack (top at right): J | D | N | K | O | H | G

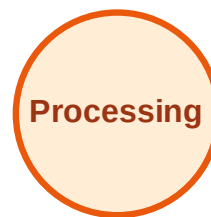
Visited: A, B, C, F



DFS Traversal

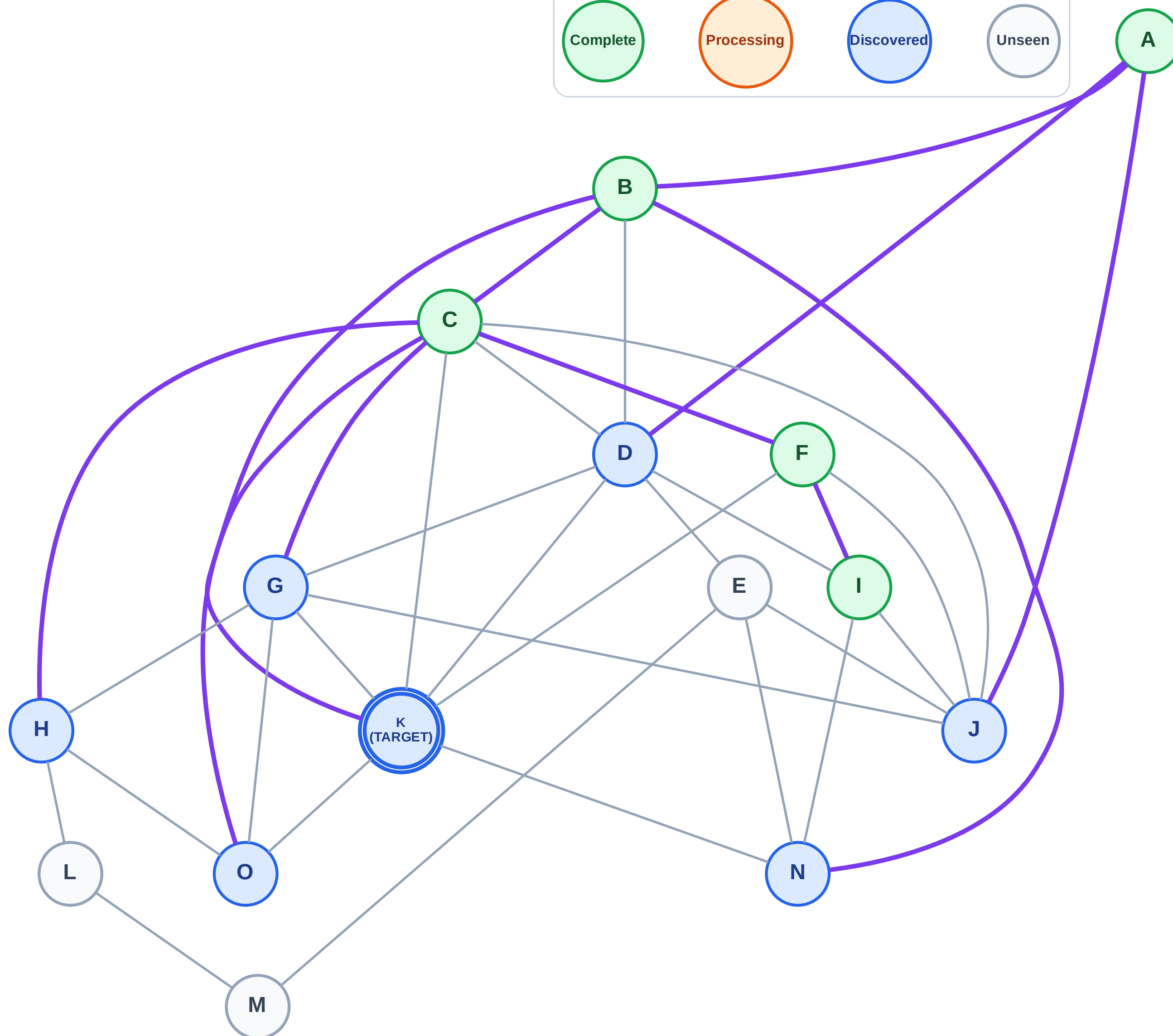
Step 11: No new neighbors from I

Legend



Stack (top at right): J | D | N | K | O | H | G

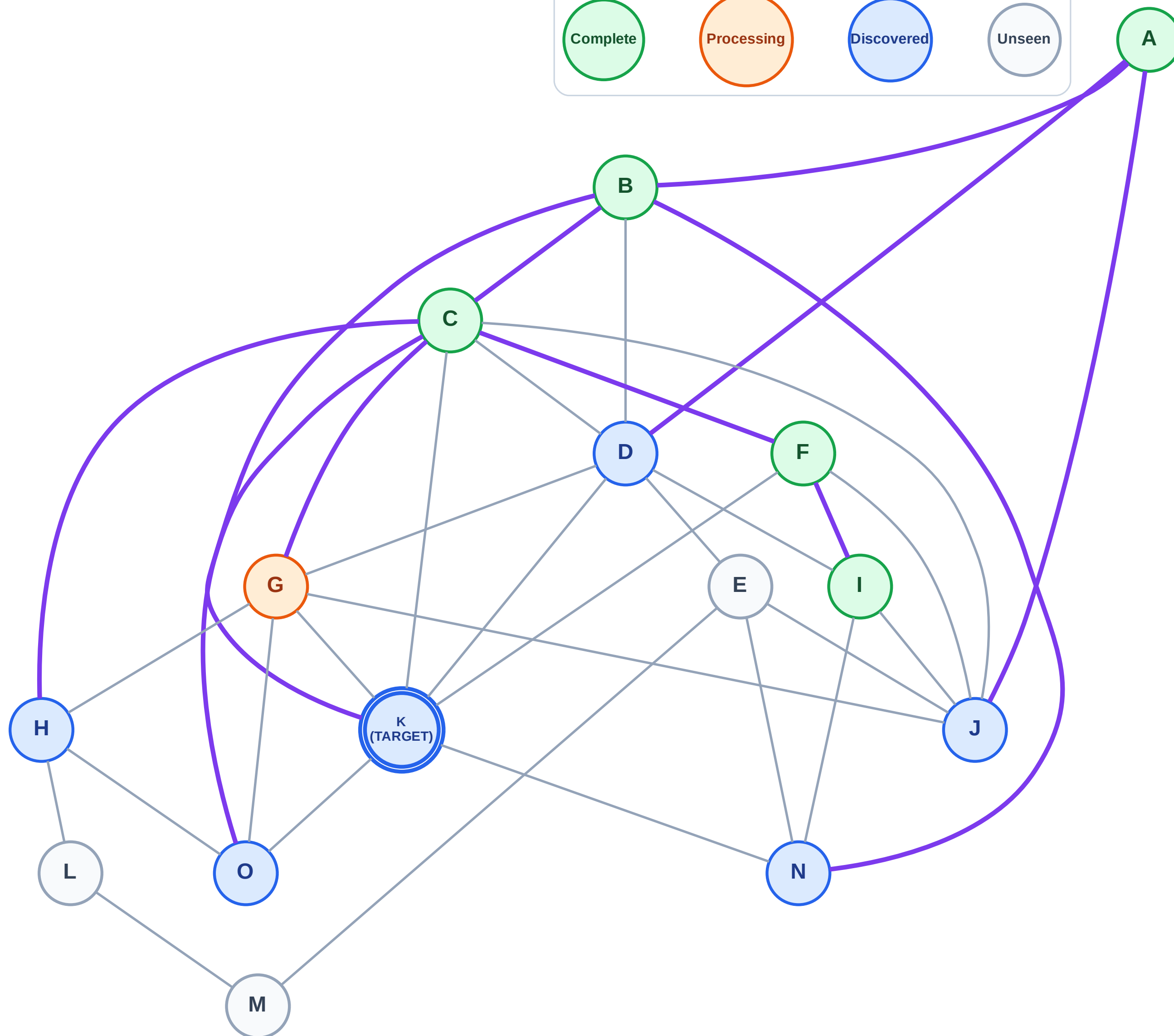
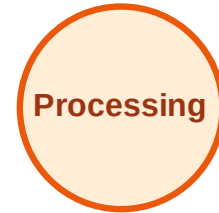
Visited: A, B, C, F, I



DFS Traversal

Step 12: Process node G

Legend



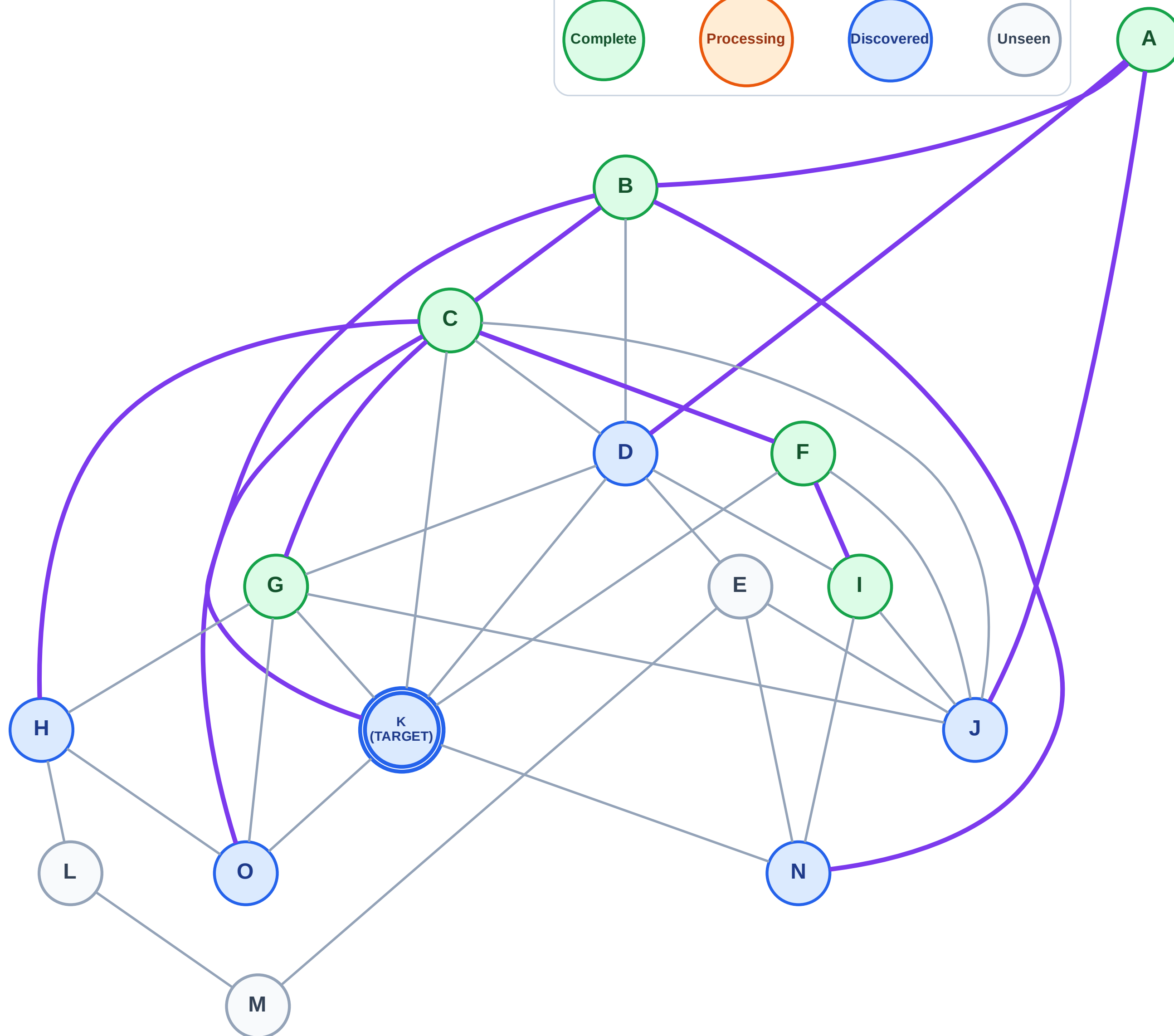
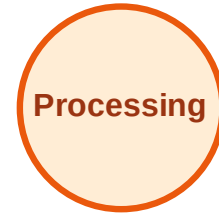
Stack (top at right): J | D | N | K | O | H

Visited: A, B, C, F, I

DFS Traversal

Step 13: No new neighbors from G

Legend



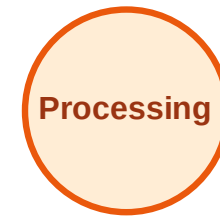
Stack (top at right): J | D | N | K | O | H

Visited: A, B, C, F, G, I

DFS Traversal

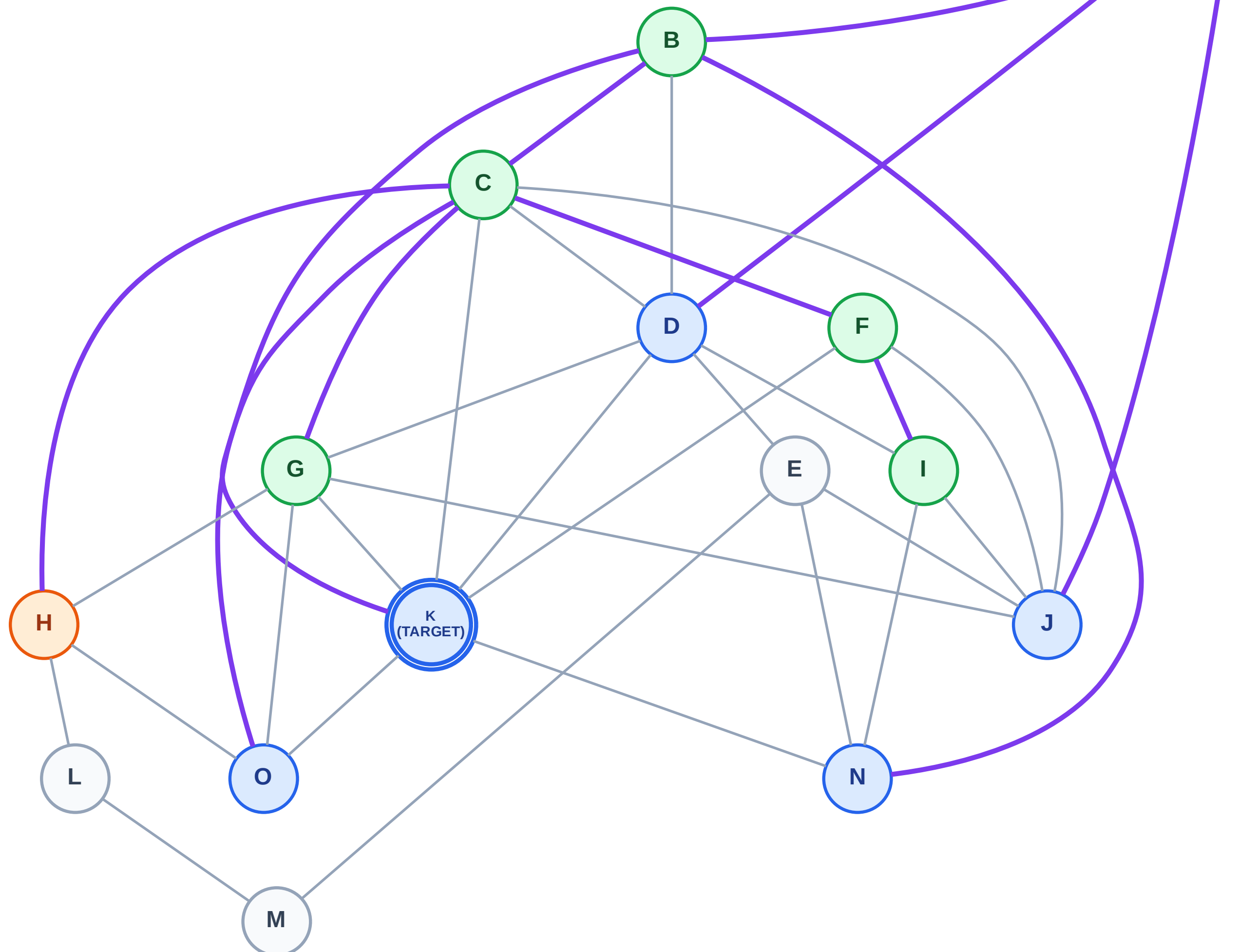
Step 14: Process node H

Legend



Stack (top at right): J | D | N | K | O

Visited: A, B, C, F, G, I



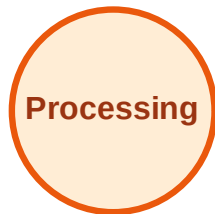
DFS Traversal

Step 15: Discover L from H

Legend



Complete



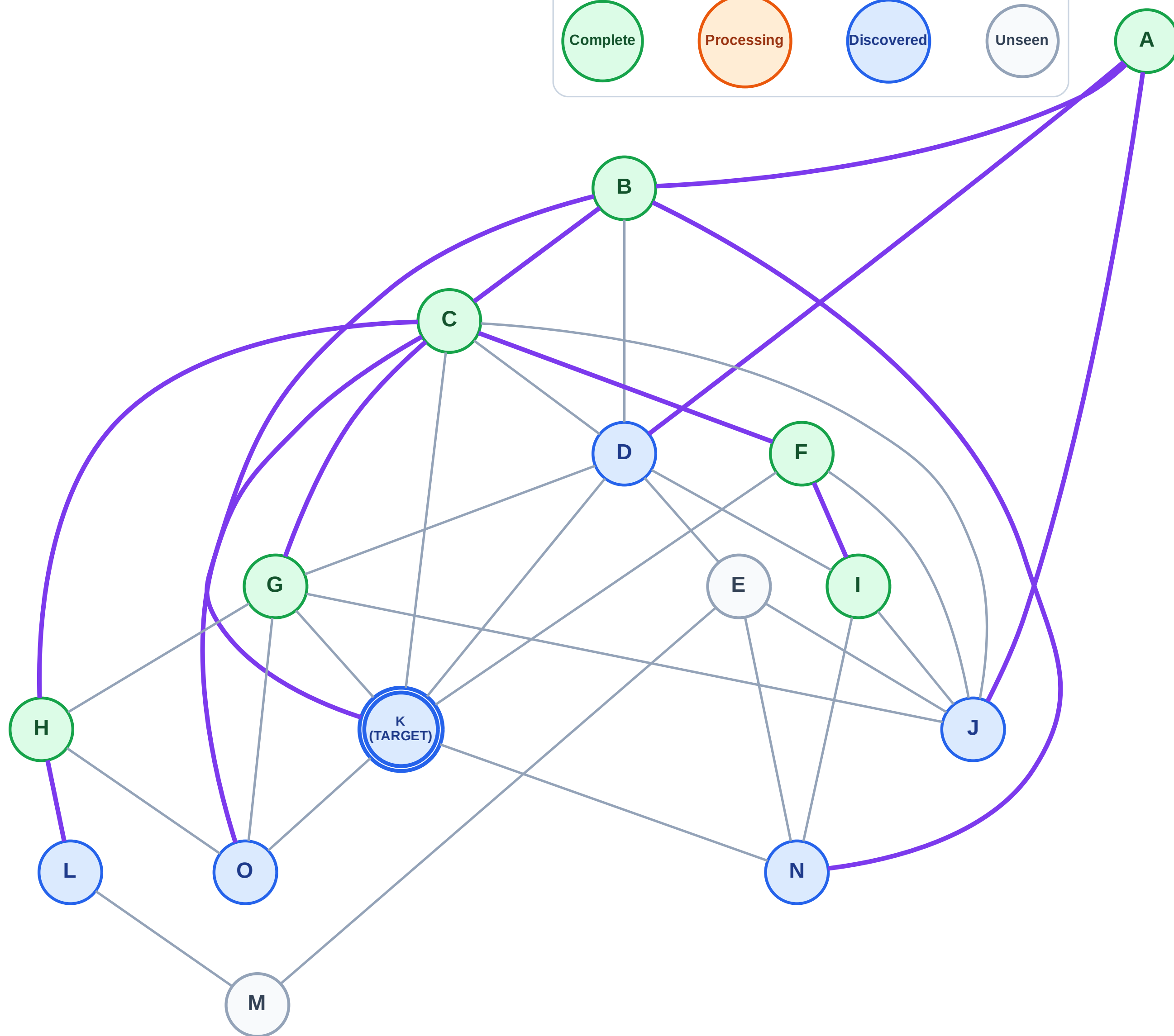
Processing



Discovered



Unseen

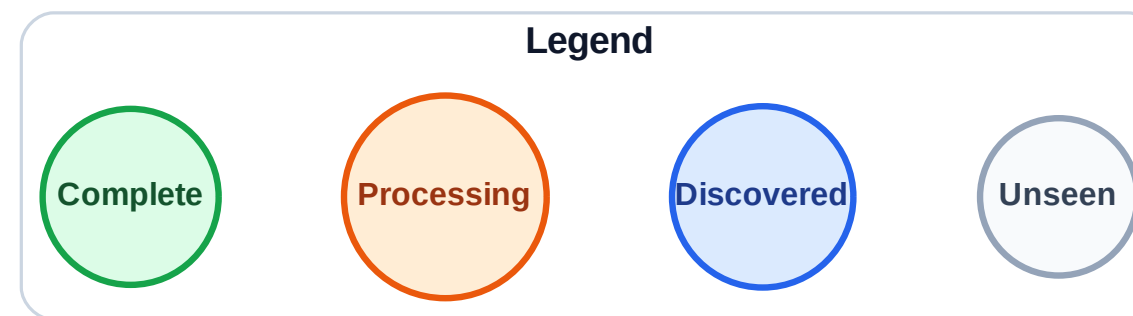


Stack (top at right): J | D | N | K | O | L

Visited: A, B, C, F, G, H, I

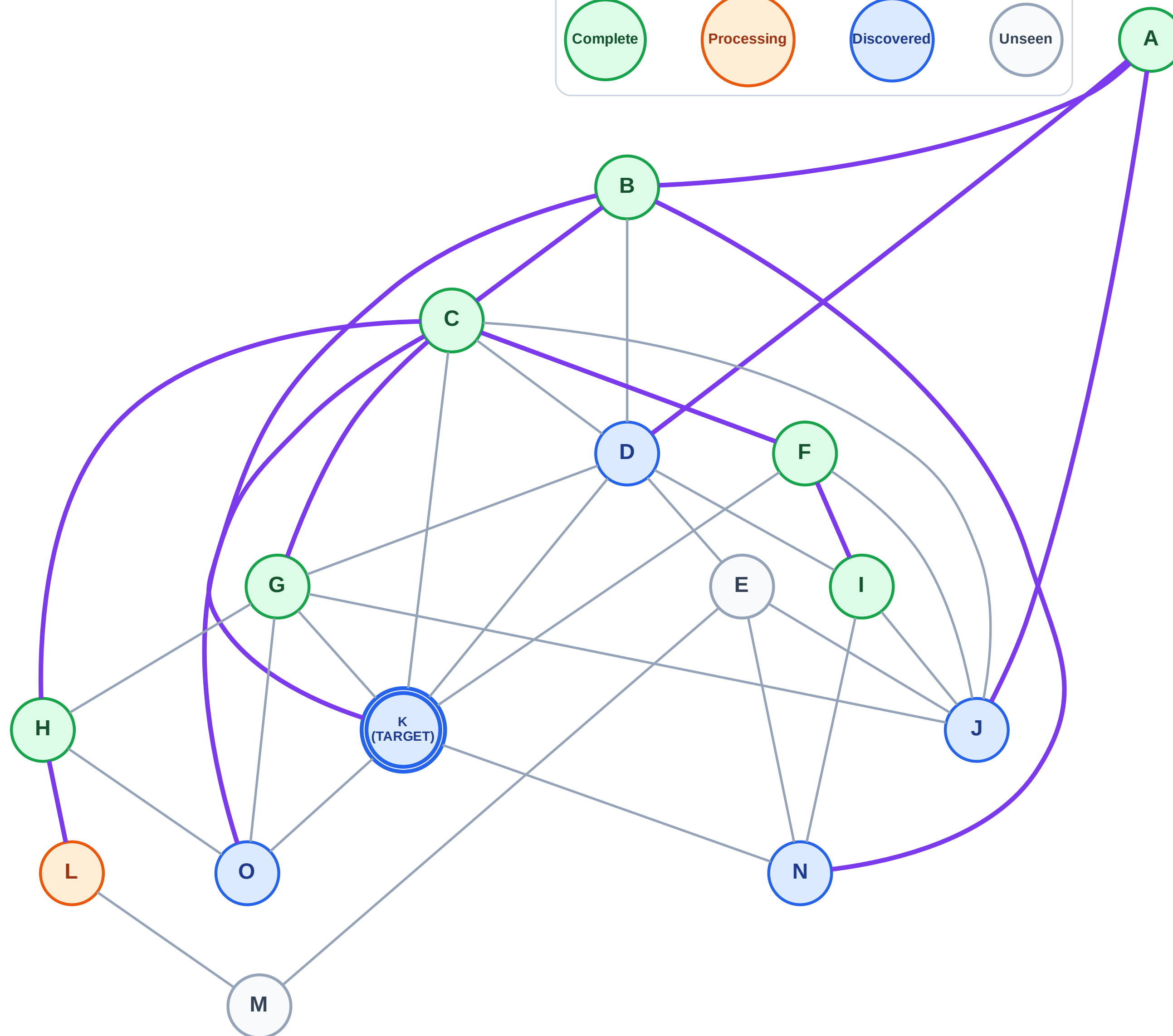
DFS Traversal

Step 16: Process node L



Stack (top at right): J | D | N | K | O

Visited: A, B, C, F, G, H, I



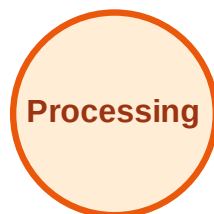
DFS Traversal

Step 17: Discover M from L

Legend



Complete



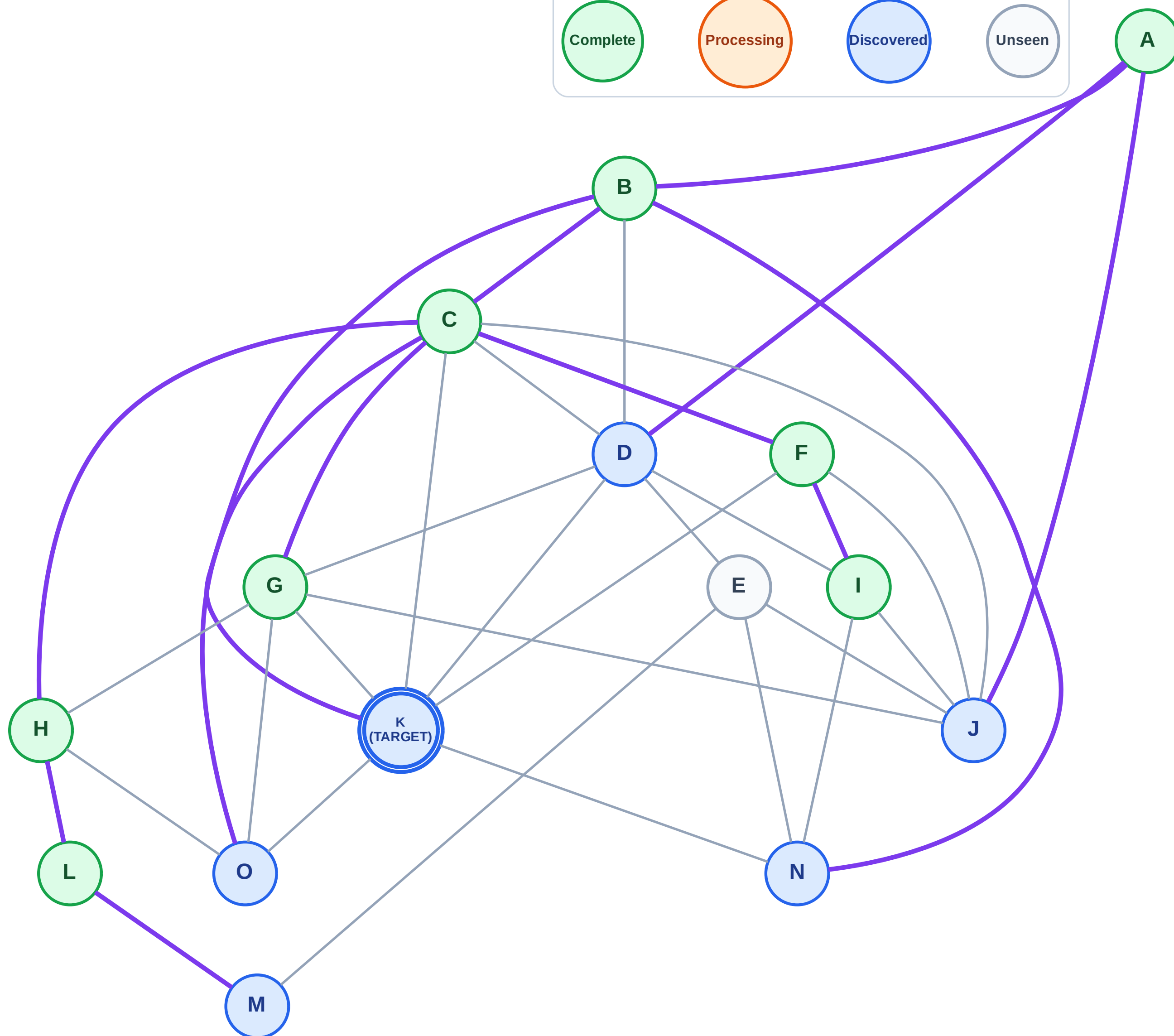
Processing



Discovered



Unseen



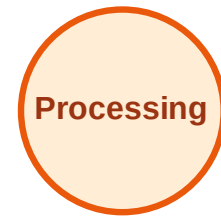
Stack (top at right): J | D | N | K | O | M

Visited: A, B, C, F, G, H, I, L

DFS Traversal

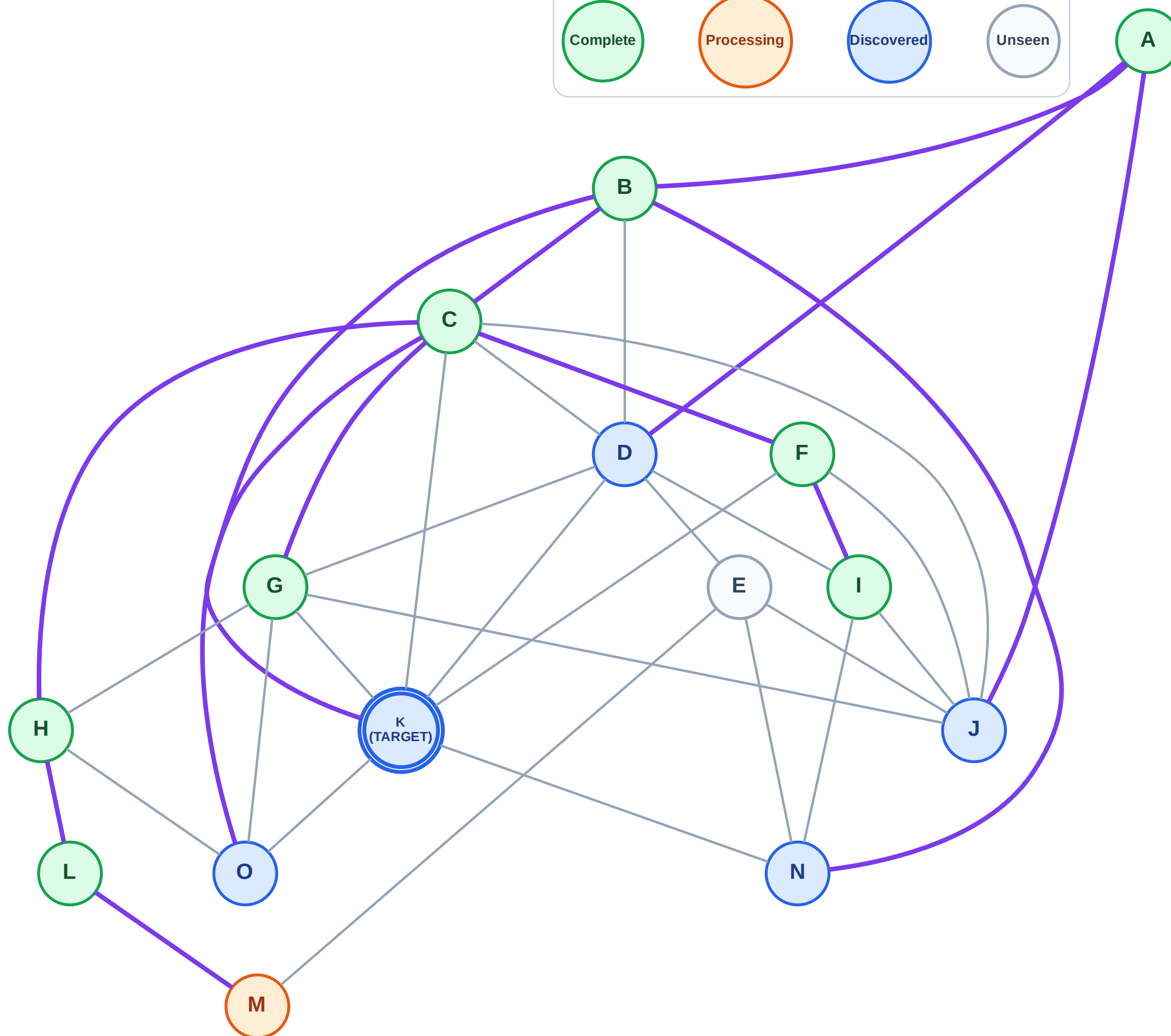
Step 18: Process node M

Legend



Stack (top at right): J | D | N | K | O

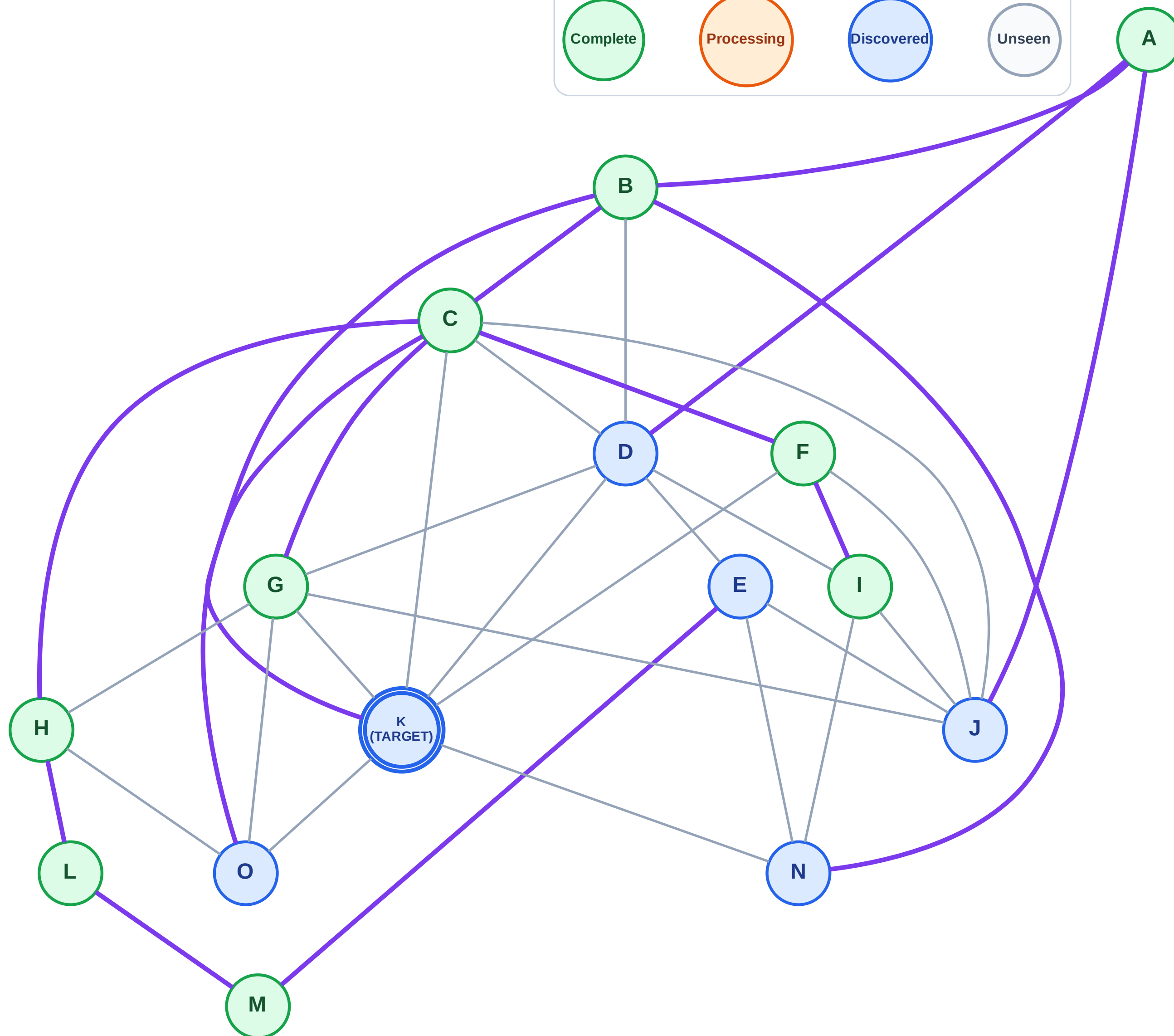
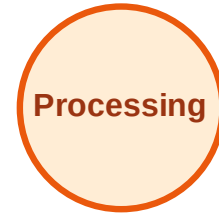
Visited: A, B, C, F, G, H, I, L



DFS Traversal

Step 19: Discover E from M

Legend



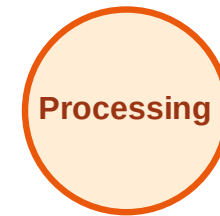
Stack (top at right): J | D | N | K | O | E

Visited: A, B, C, F, G, H, I, L, M

DFS Traversal

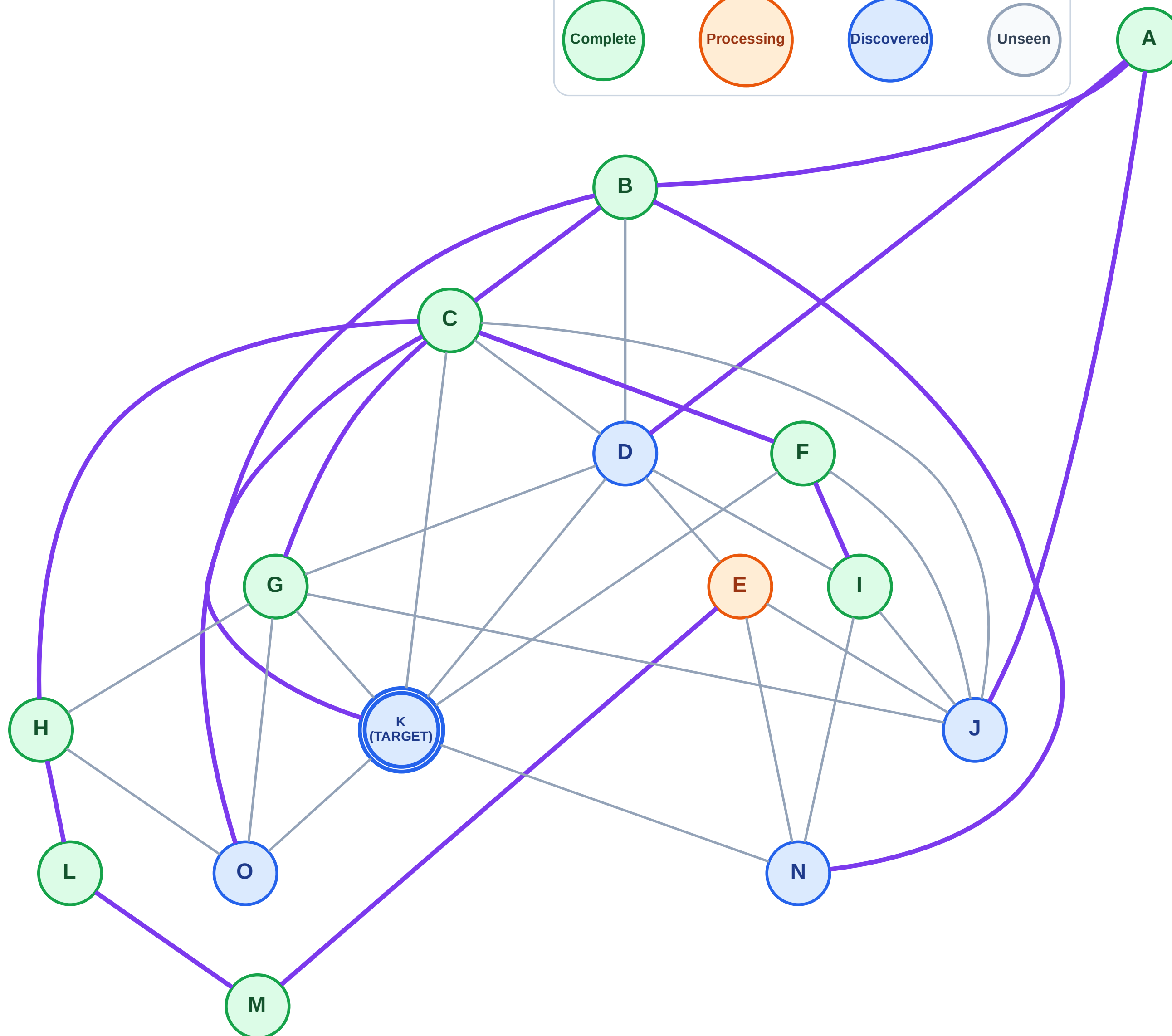
Step 20: Process node E

Legend



Stack (top at right): J | D | N | K | O

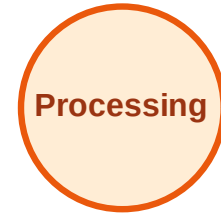
Visited: A, B, C, F, G, H, I, L, M



DFS Traversal

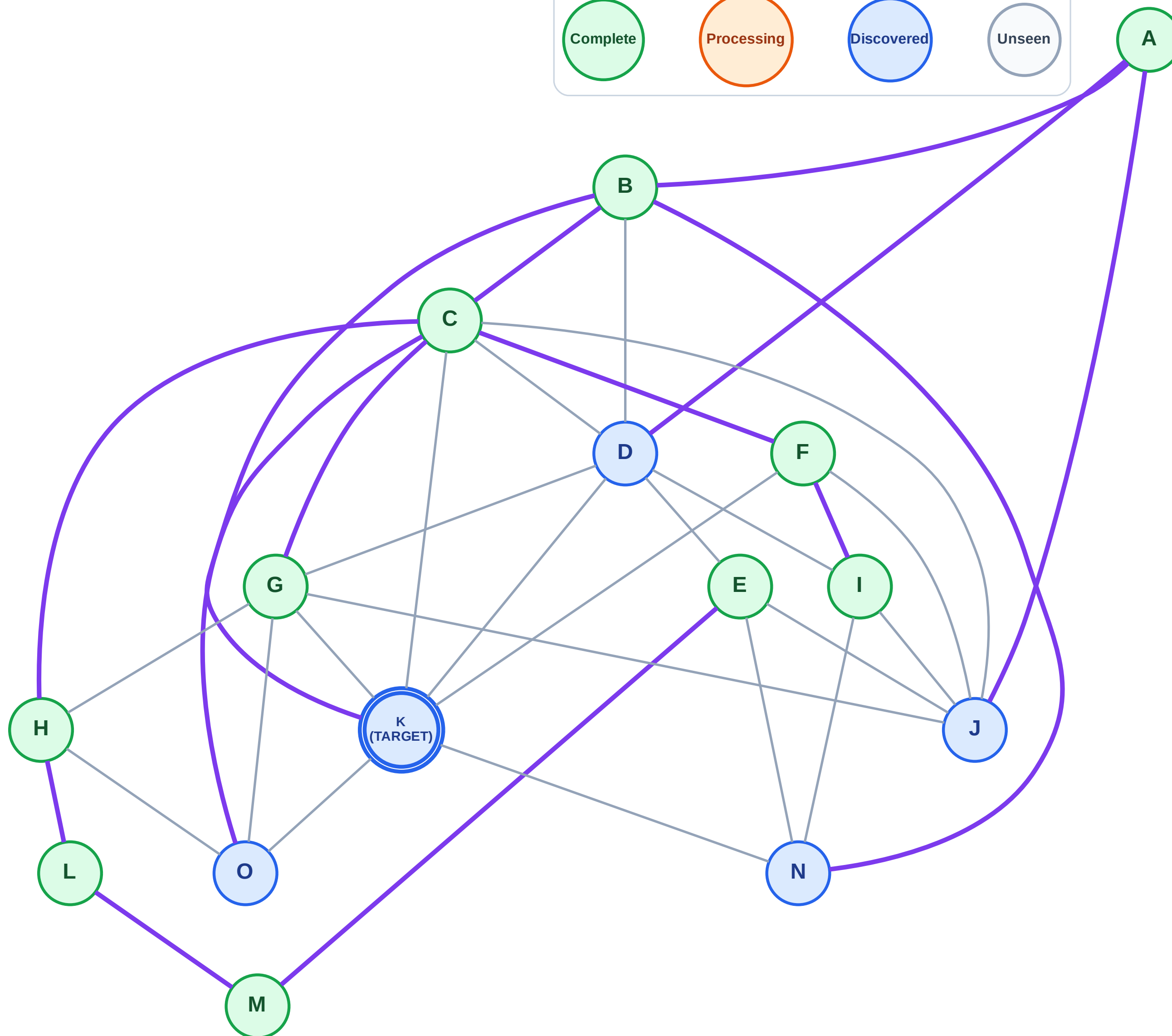
Step 21: No new neighbors from E

Legend



Stack (top at right): J | D | N | K | O

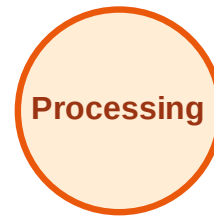
Visited: A, B, C, E, F, G, H, I, L, M



DFS Traversal

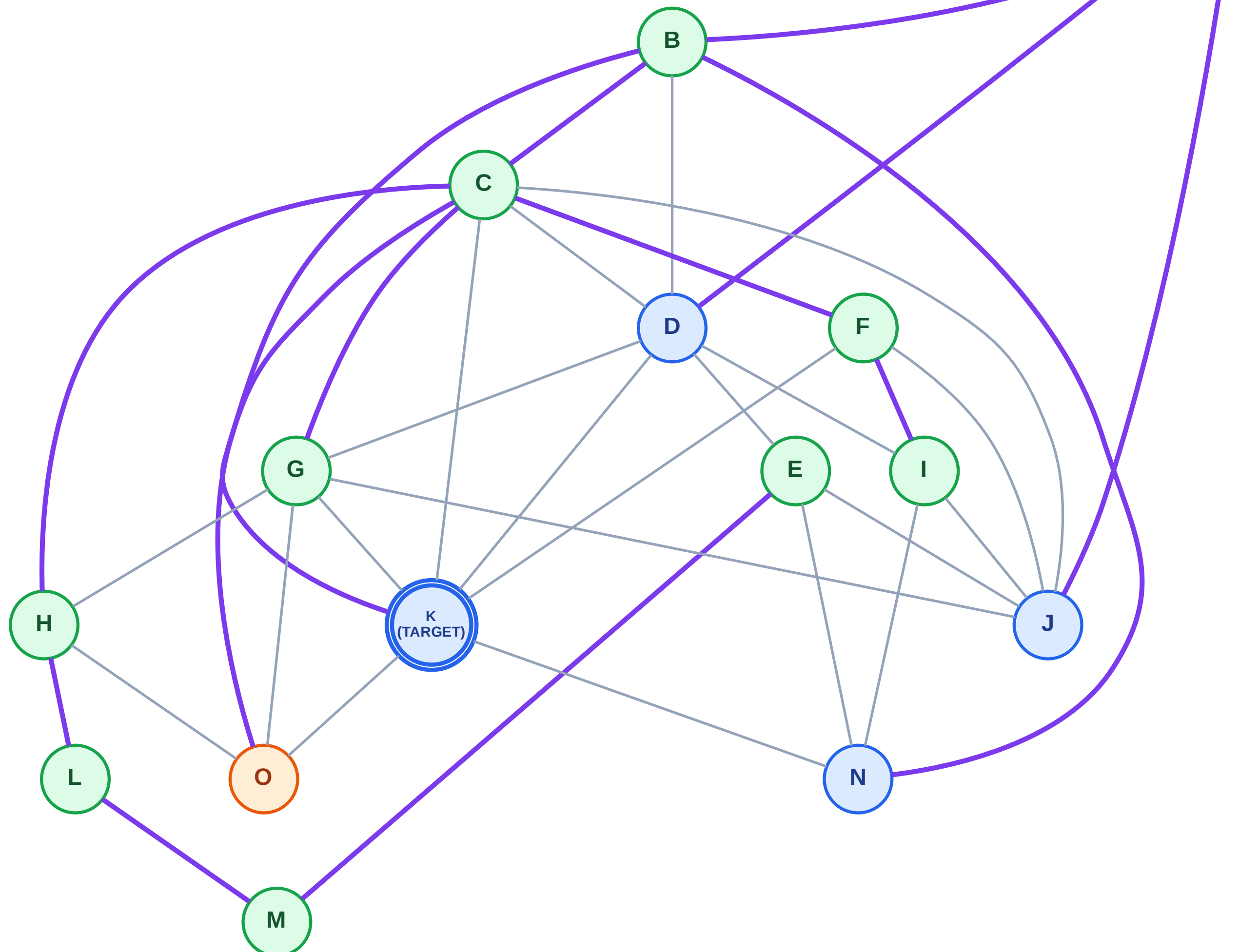
Step 22: Process node O

Legend



Stack (top at right): J | D | N | K

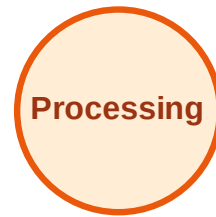
Visited: A, B, C, E, F, G, H, I, L, M



DFS Traversal

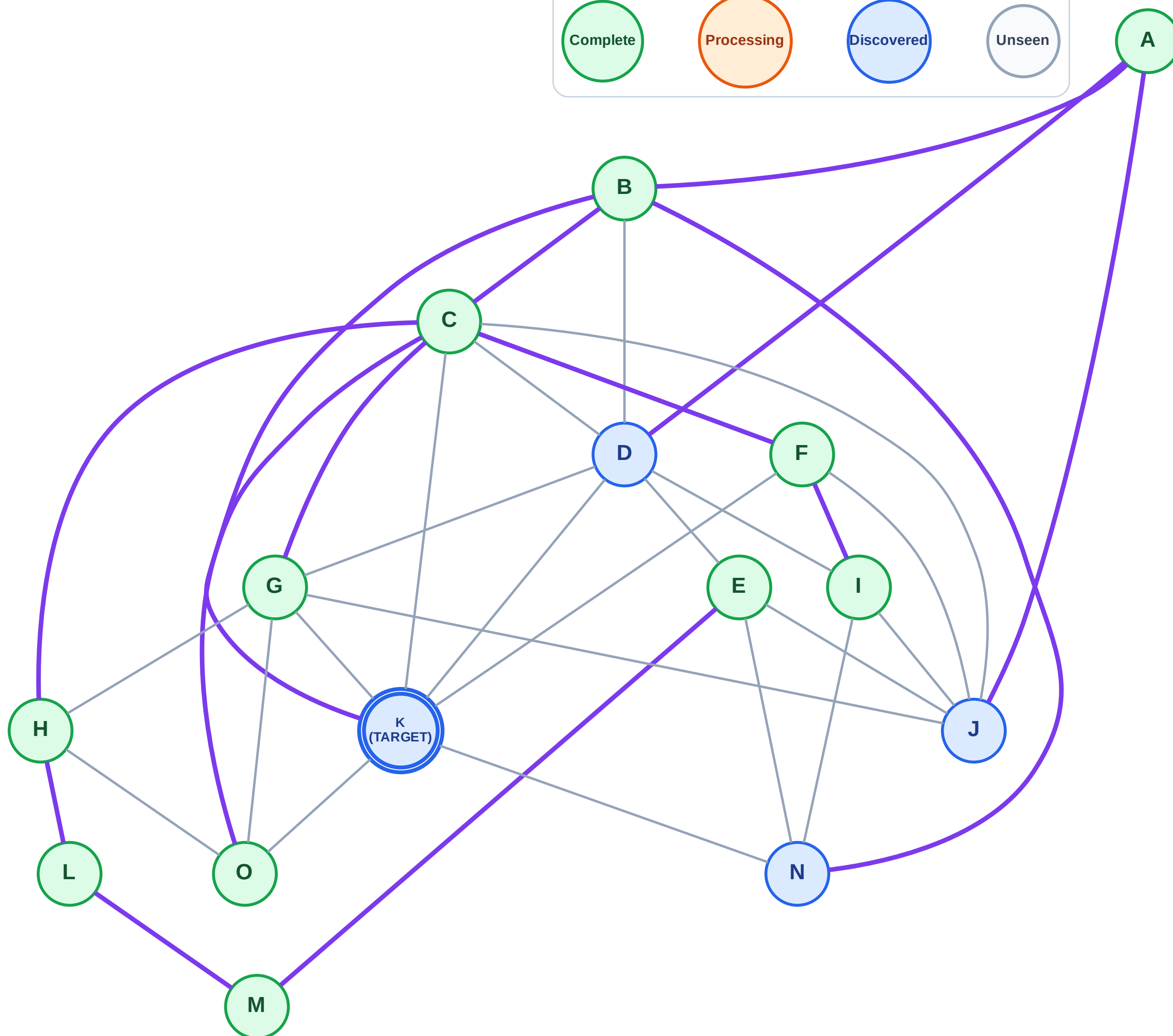
Step 23: No new neighbors from O

Legend



Stack (top at right): J | D | N | K

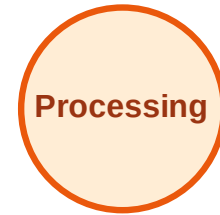
Visited: A, B, C, E, F, G, H, I, L, M, O



DFS Traversal

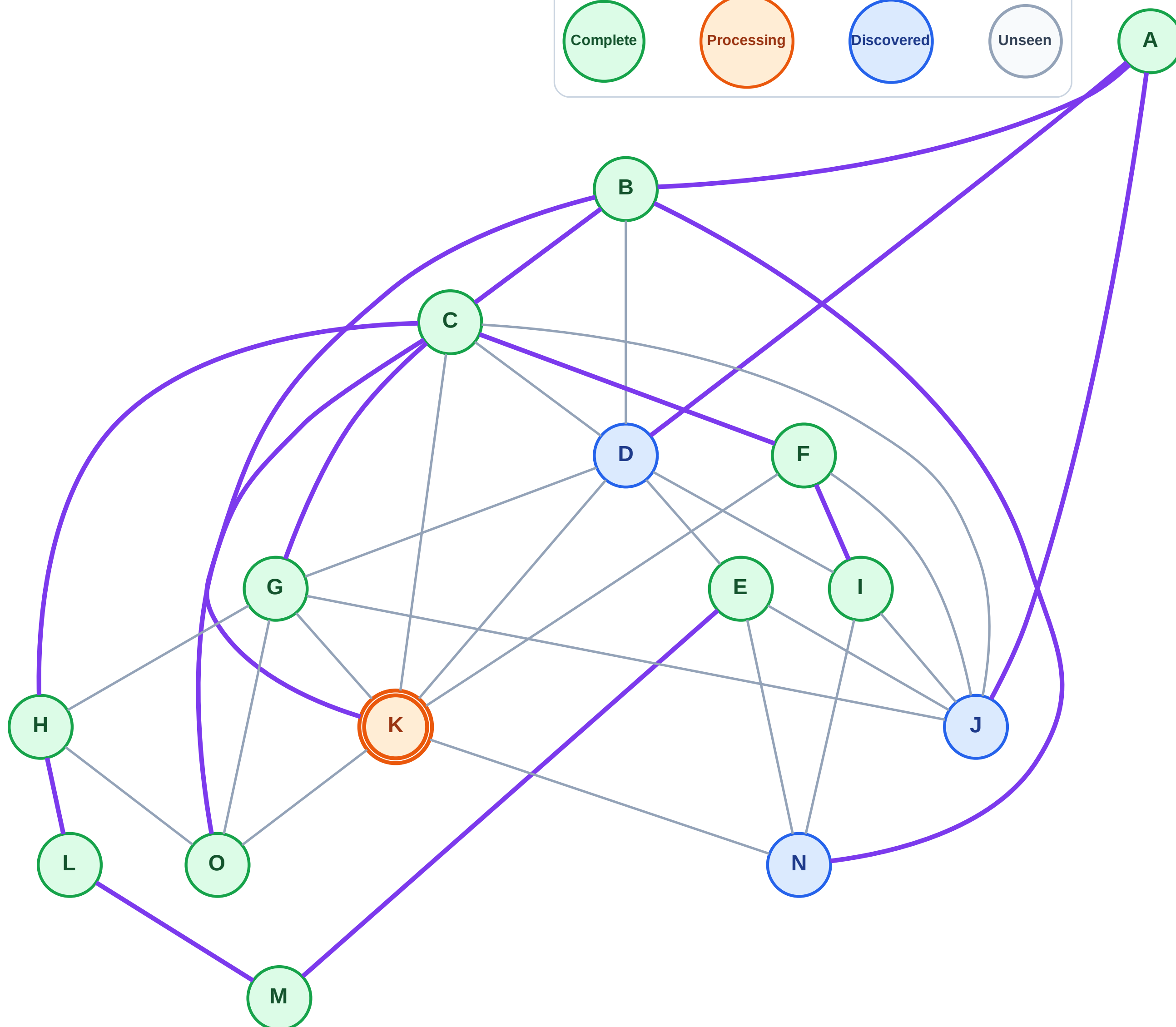
Step 24: Process node K

Legend



Stack (top at right): J | D | N

Visited: A, B, C, E, F, G, H, I, L, M, O



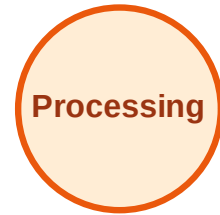
DFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): J | D | N

Visited: A, B, C, E, F, G, H, I, K, L, M, O

